Panasonic

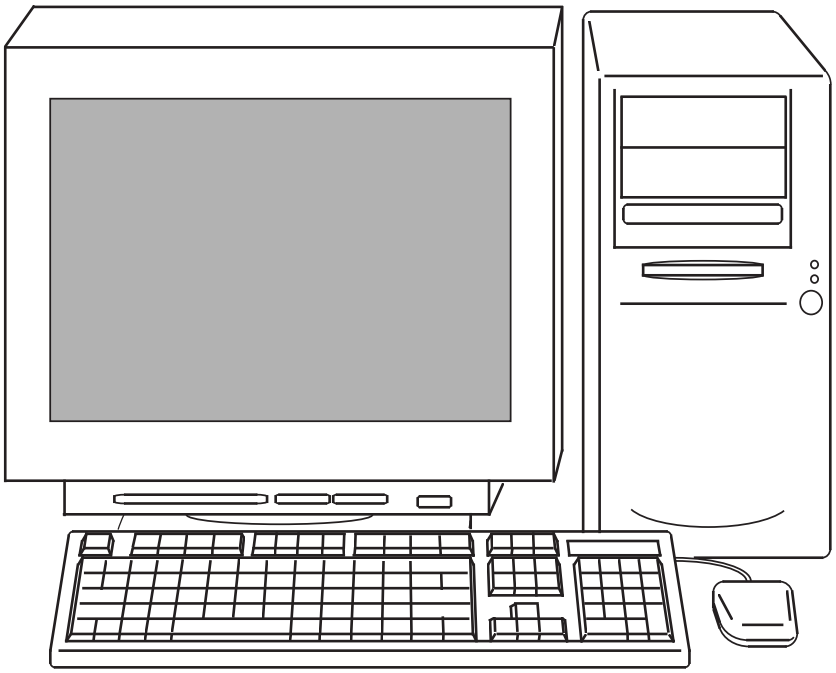
Operating Instructions

Component Supply Navigator

MES for Electronics Assembly

PanaCIM-EE

Model No. **NM-EJS3B**



Thank you for purchasing “Panasonic Product”.
- Read the instruction manuals carefully and use the product safely.
- Read “Safety Notes” before you start using the product.
- Store this instruction manual in a safe place.

N7201A633E06 (V08.08)
201712

Original Instructions

- Use of this machine:
  This machine is designed and constructed to be used for placing electric and electronic components on PCB.
  This machine shall be used for this purpose only.
  It shall not be used for any other purpose.

- Operation of this machine:
  This machine shall be operated by professional, specially trained operators.
  It shall not be operated by non-professional, not specially trained operators.

  For information regarding available training courses, please contact Panasonic Smart Factory Solutions Co., Ltd., Sales company or Sales agent.

⚠ Warning

Operate the machine after receiving special training



Risk of injury due to improper operation.

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About the Instruction

■ Operating instruction contents

The following operating instructions are included in the DVD-ROM attached to this machine. The instructions are divided according to work or function. Use one that is appropriate for the purchased module (application) or your purpose.

∗ BM series machine is included in the explanation of this manual, but its functions are not supported.

Type	Description
<General>	
Introduction	Overview of machine and basic setting/operation/analysis/utility.
Basic Operation (PanaCIM Web Application)	PanaCIM Web application's basic operation.
<Modules>	
Production Analysis	Collect and analyze machine's running information and display its statistics on real time basis to support productivity improvements (recognition/pickup error rate improvement, machine trouble stop time reduction) for mounting line.
Traceability	Combine PCB information read by a barcode reader on machine or conveyor and production record or material/part information of machine to manage history of PCBs for quality control and market support.
Material Verification	Verify material/part information and recipes while setting up material/parts in production setup/model change/splicing/part exhaust to prevent wrong material/part set up.
Product Changeover and Control	Auto download recipe and adjust lane width based on PCB ID for quicker model change.
Production Monitoring	Retrieve production information and operation information on real time basis and display line status on the monitor and PDA to provide operators with instructions for events such as error or part exhaust. This is to reduce mount line stop time or to prepare in advance for better productivity.
Enterprise Link	A function to output the data collected from machines during production (material verification data, trace data, operation performance data) in the standard format so external applications can reuse the data. Part information can be loaded also in the standard format.
Material Control	A function to provide the location and the number of parts or feeders stocked in the production or preparation area. MSD function controls MSD parts' expired date.
Component Supply Navigator	The Component Supply Navigator optimizes the supply sequence to provide a shortest route of an operator without component shortage. It can prevent short time breakdown in the production line occurred due to component shortage and contribute to manpower saving.
Maintenance	For the purpose of reducing time spent on maintenance planning / control operations, asset (machines, heads and feeders) profiles can be controlled on the floor, or the maintenance history of each asset can be recorded.

License CD names for modules are as follows.

Module Name	License CD Name
Production Analysis	Production Analysis
Traceability	Traceability
Material Verification	Material Verification
Product Changeover and Control	Product Changeover and Control
Production Monitoring	Production Monitoring & Dispatch
Enterprise Link	Enterprise Link
Material Control Base	Material Control Base
Material Control Allocation	Material Control Allocation
Material Control MSD	Material Control MSD
Component Supply Navigator	Component Supply Navigator
Maintenance	Maintenance

- This manual describes how to use this function with the machines for NPM series, AM series or CM/DT series. In this manual, ‘the machine’ or ‘this machine’ indicates the machine for NPM series, AM series or CM/DT series.
- When this function is used with machine for NPM series, AM series or CM/DT series, the separate manual for the machine besides this manual is available. If you do not have it on hand, contact us.
- “Mobile scanner” and “Wireless scanner” have the same meaning in this manual.

Terms

Here is a term used in this instruction manual.
As for PanaCIM basic terms, refer to 'PanaCIM-EE Instruction Series: Base.'

Terms	Description
Line Side Parts Rack	Parts rack placed near the mounting line. Generally, for improvement of the operation ratio, parts used in the line are stored near the line in advance.

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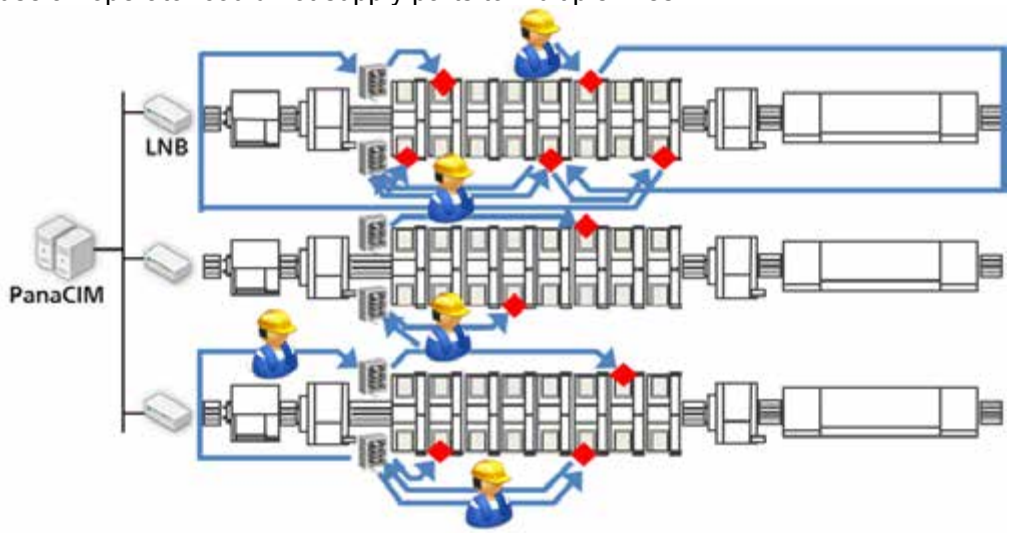
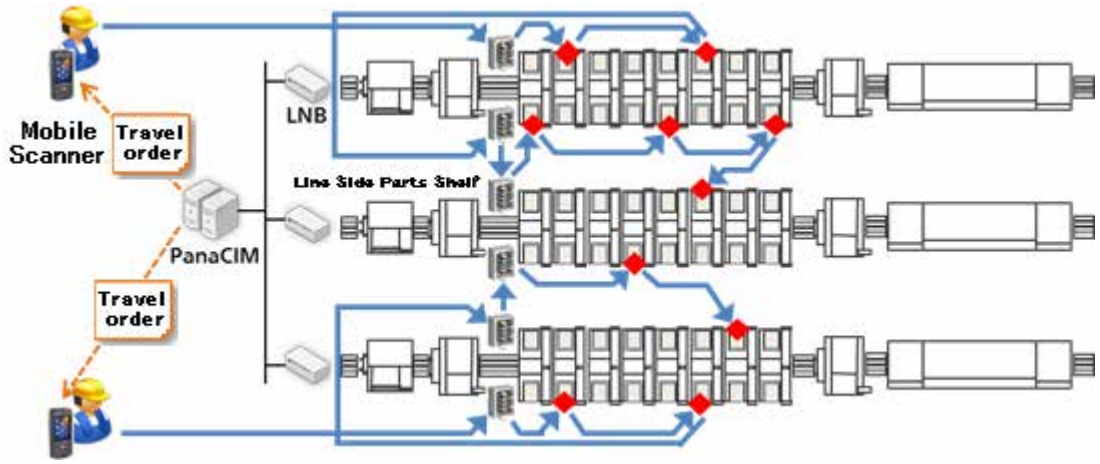
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1. Preparations

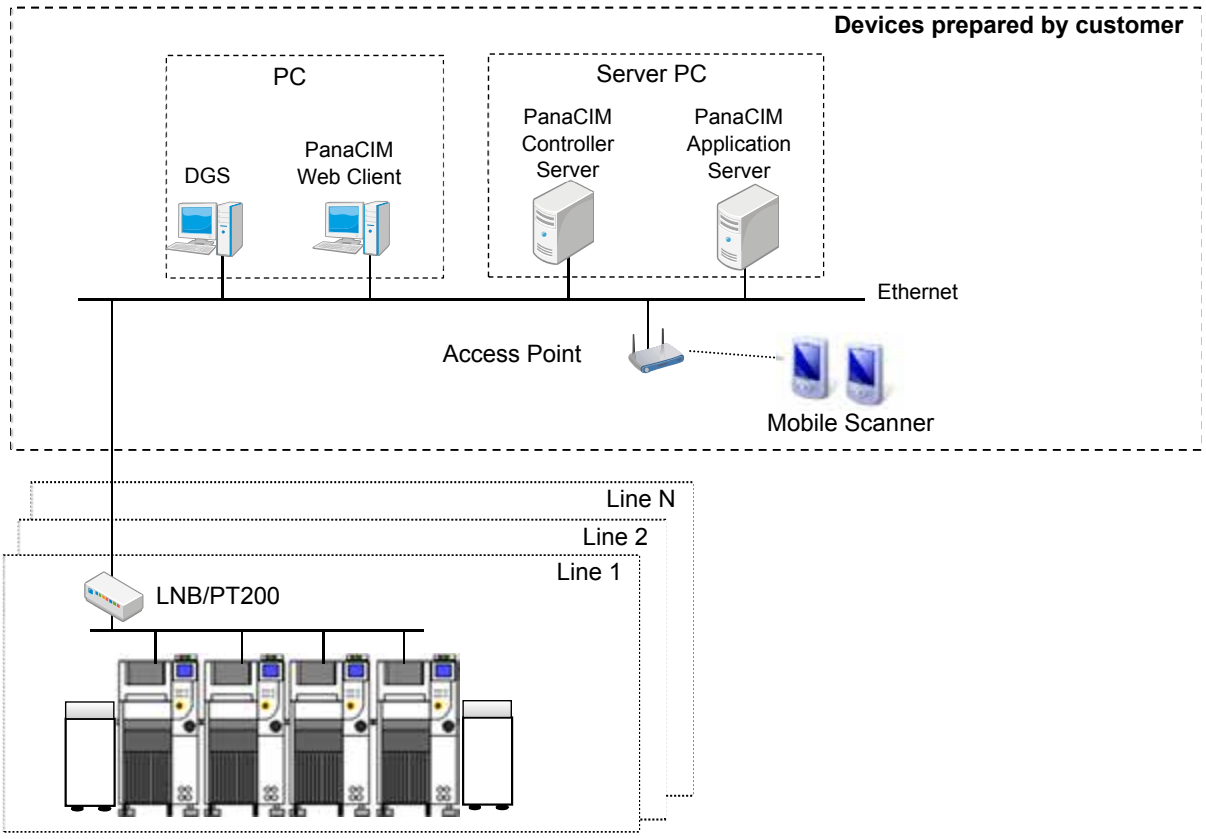
1.1 Features

- This product aims to reduce the time and personnel needed for part supply works such as splicing tapes or replacing trays.
- Group the parts that are expected to exhaust within a certain range of time, and optimize the route on real time basis for operators to travel the shortest distance to supply the part groups.
- Send a part supply order with the each operators' optimized route to operator' mobile scanner.
- Manage the information of multiple lines together and order part supply that involves multiple lines.

Before	Unncessary travelling (back and forward the same route multiple times) reduced work efficiency and operation rate (due to supply delay or momentals stops). Old system did not allow checking other lines' information so more operators needed beacuse an operator could not supply parts to multiple lines. 
After	Parts can be supplied without unnecessary travelling by following the order, and the time for supply patrol work will be shorter. One operator can supply parts to multiple lines so less personnel needed. 

1.2 System Configuration

The following system configuration is needed for this product.



■ **Devices prepared by customer**

PC	DGS	Used to generate recipes. See [Data Generation System Manual] for the computer's hardware requirements. ∗ Used only in LNB line
	PanaCIM Web Client	Used for monitoring PanaCIM settings and supply work progress. See [PanaCIM-EE Manaal (Installation)] for the computer's hardware requirements.
	PT200	Used to generate recipes. ∗ Used only in CM/DT line
Server PC	PanaCIM Controller Server	Used as PanaCIM controller server. See [PanaCIM-EE Manaal (Installation)] for the computer's hardware requirements.
	PanaCIM Application Server	Used as PanaCIM application server. See [PanaCIM-EE Manaal (Installation)] for the computer's hardware requirements.

Mobile Scanner	Used to check part verification and part supply order. Use Motorola [MC55] or [MC70] or [MC75]. See [PanaCIM-EE Manaal (Installation)] for these devices' requirements.
Access Point	Used to establish communication between PanaCIM and mobile scanner. See [PanaCIM-EE Manaal (Installation)] for these devices' specifications.

1.3 List of Functions

This product has the following functions.

Part Supply Patrol Order

- Supply Route Optimization — Group the parts that are expected to exhaust within a certain range of time, and optimize the route on real time basis for operators to travel the shortest distance to supply the part groups.
- Show Travel Order — Show the list of part groups in the order of priority that the scanner's log-in operator should supply on mobile scanner screen.

Supply Work Progress Monitoring

- Work Progress Monitor — Show the list of part supply patrol orders sent to operators on web browser. Supplied parts are deleted on real time so the work progress of the operators can be monitored.

Coordination with Matrial Management Module

With the license of [Matrial Management (Basic)] and [Material Management (Allowance)], the following functions will be enabled.

- Part Location — Shows the presence of parts at the parts shelf on the side of lines on the mobile scanner screen.
- Delivery Order to In-line Shelf — Show the number of reels (trays) of the parts being used in production stoked in the line side shelf on the computers in the kitting area so operators know what and when to deliver to the lines.

Links with Production Analysis

The following function will be available after you introduce [Production Analysis License].

- Supply Work Record Management — Show the supply work results of individual operators that were done according to the travel order.

1.4 Limiting Conditions

The following shows compatible machines and non-compatible supply units.

1.4.1 Compatible Machine

The following machines are supported.

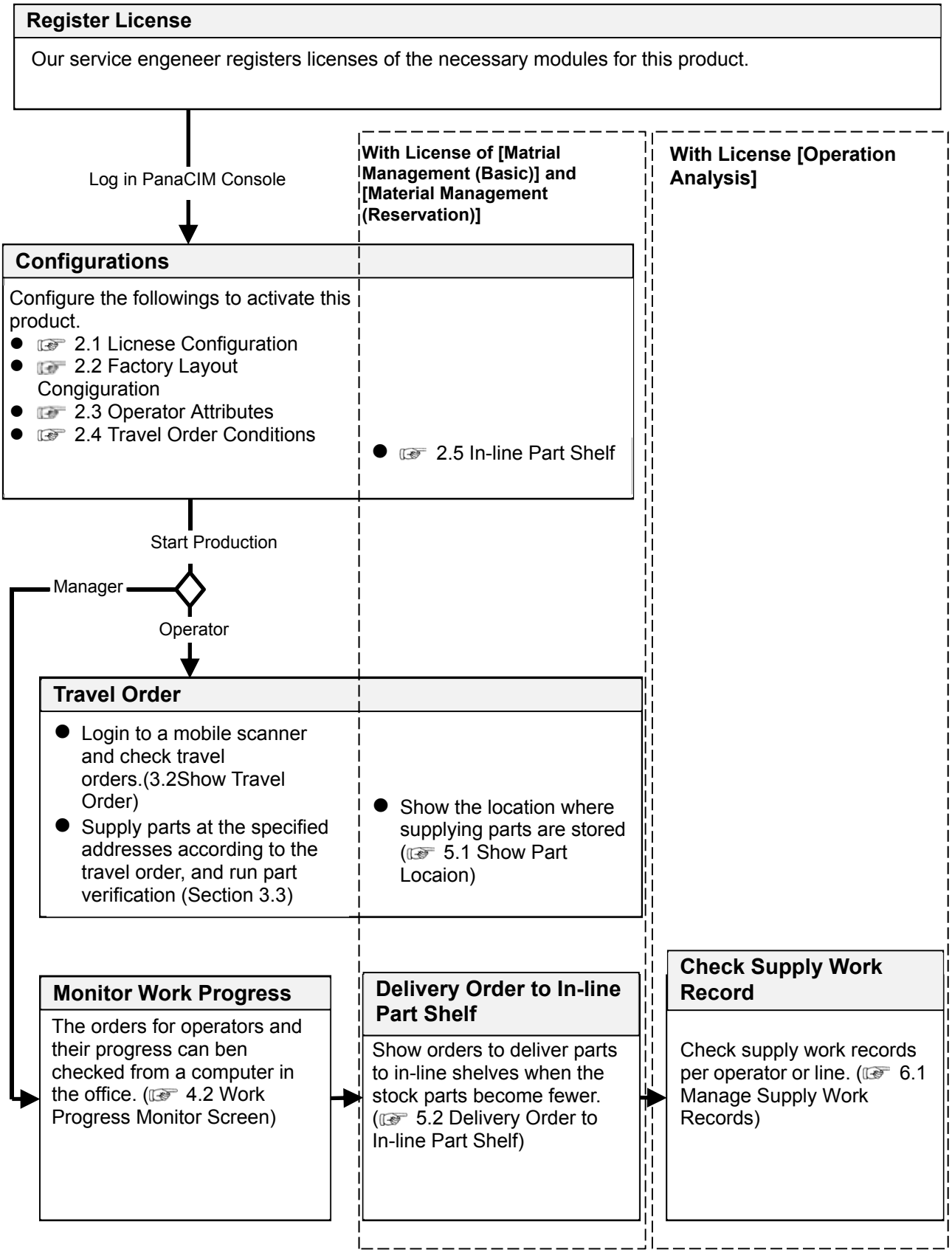
- NPM X series: NPM-DX
- NPM series: NPM, NPM-D, NPM-D2, NPM-D3, NPM-W, NPM-W2, NPM-TT, NPM-TT2
- AM100
- CM/DT series: CM602-L, CM232-M, CM212-M, CM101-D, CM402-L, CM402-M, CM401-L, CM401-M, DT401-F, DT401-M
  ∗ In CM/DT series, trays are not supported

1.4.2 Non-compatible Supply Unit

Supply instructions cannot be provided for the supply units below that do not have the remaining quantity control function.

- Intelligent stick feeder
- Manual setting tray

1.5 Flow of Operations



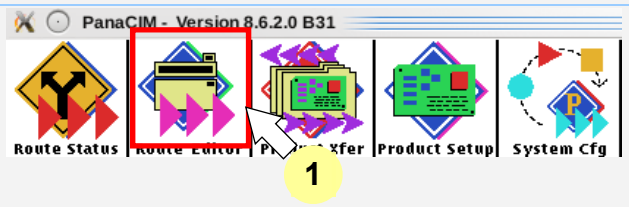
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2. Configuration

2.1 License Configuration

To use Parts Supply Navi, the license needs to be enabled from Route Editor on PanaCIM controller. Stop the machine before you start the following steps.

1
Go to PanaCIM Controller's application tool bar, click <Route Editor>.
* See "PanaCIM-EE Introduction 2.1.2 How to Start" for how to start the application tool bar. .

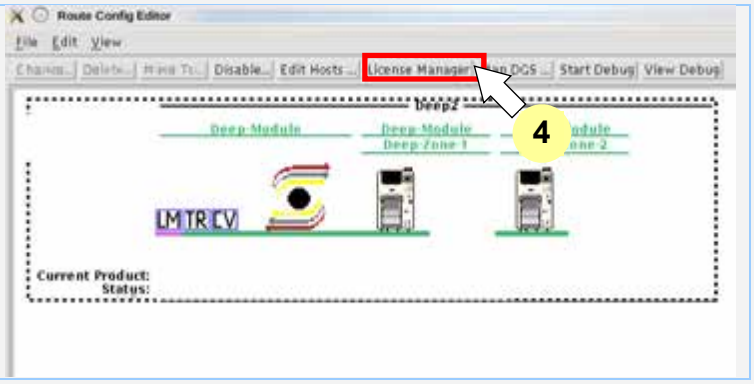


2
On <Route Config Editor> screen, select a route.

Click <Disable>.

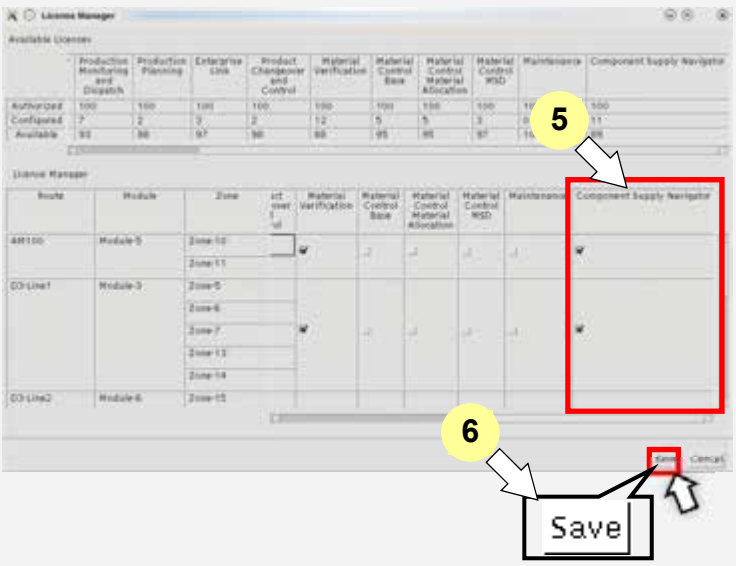


4
Click <License Manager> next.



5
On <License Manager> screen "License Manager" table, check [Parts Supply Navi] for each module.

6
All configurations are finished, click [Save] and close.

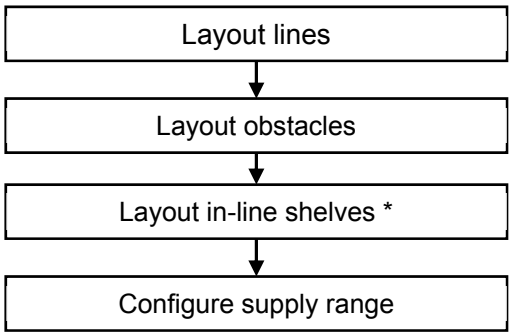


【Note】
The routes with active Components Supply Navifator/License are shown in green.
Before you change license configuration, disable the route first.

2.2 Factory Layout

Configure factory layouts such as mount line or other equipment positions that are needed for optimizing parts supply route.

The basic configuration is as follows.

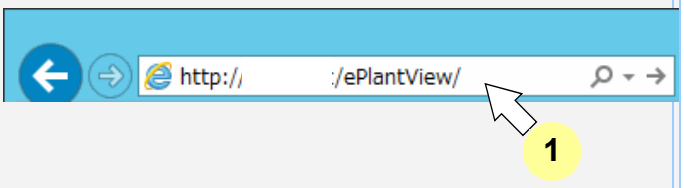


* Configurable only with valid [Material Management (Basic)] and [Material Management (Reservation)] license.

2.2.1 Start Factory Layout Designer

1
Open a Web browser and enter the following URL to access ePlantview.

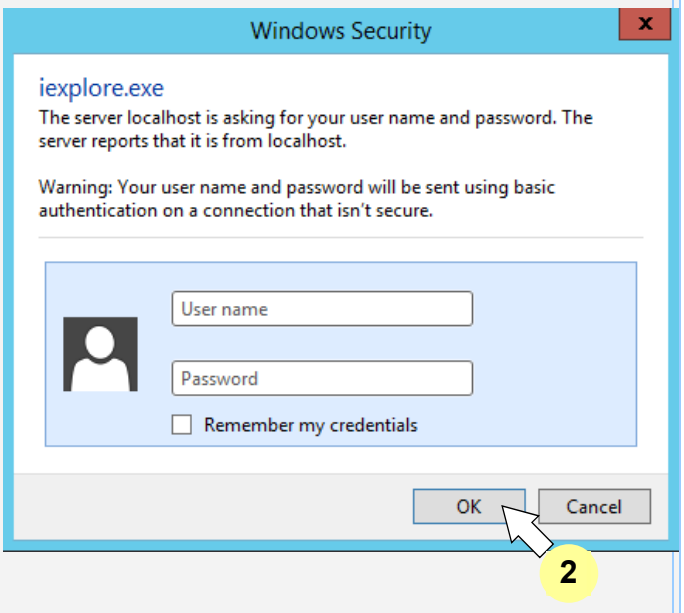
http:///(PanaCIMApplicationServerIP address)/eplantview



2
Enter login user name and password on the login screen, and click [OK].

【Note】
If [Microsoft® Silverlight] is not applied to the web browser, a pop-up window appears to install it.



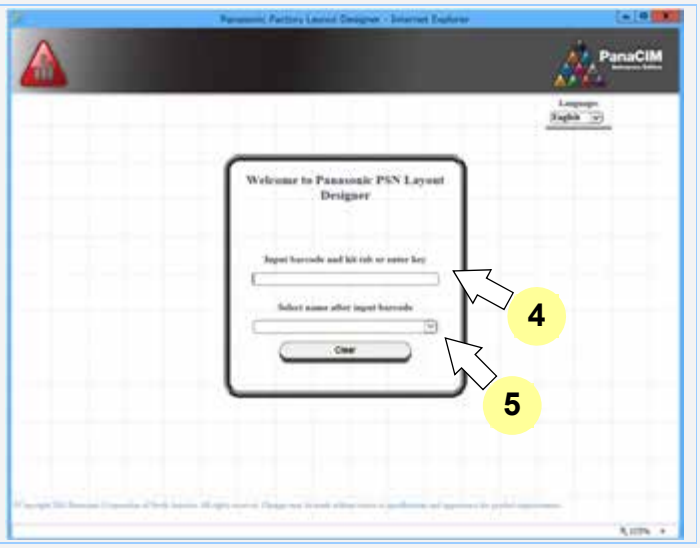


3
[PanaCIM ePlantview] menu screen appears.
Select [Administration]-[Factory Layout Designer].



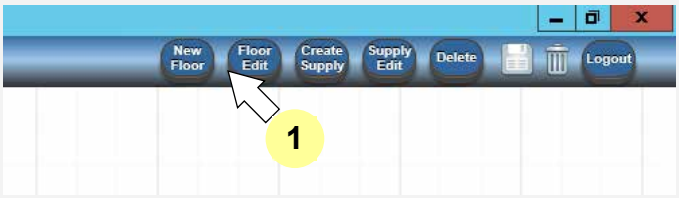
4
Factory Layout Designer starts.
Enter OperatorID barcode information and press Enter.
Select a name and login to Factory Layout Designer.

5
Select a name and login to Factory Layout Designer.



2.2.2 Line Layout

1
Create a new floor layout.
Click [New Floor] from Factory Layout Designer's menu.

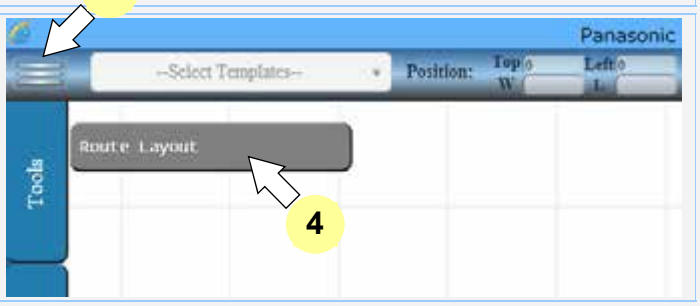


2
Make sure [New Floor] turned to red, and click [Floor Edit].

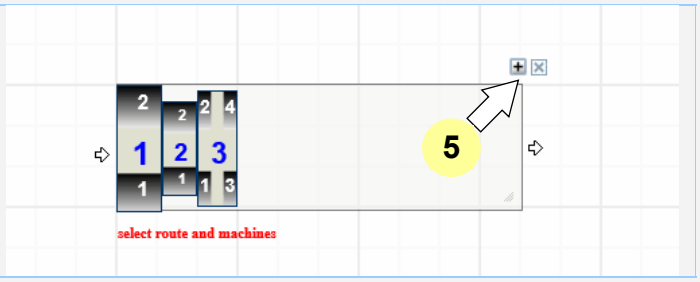


3
Click  to view side menus,

4
Click [Tool]—[Route Layout].



5
Route layout will be added.
Click [+].

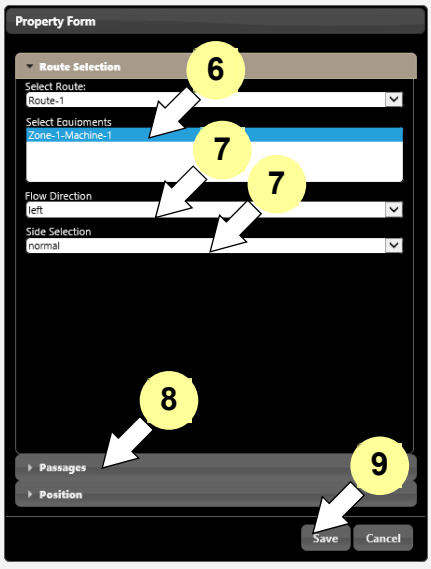


6
Select a route (line) and machines that are configured in the route.

7
Select "Flow Direction" and "Side Selection" for the machine.

8
Add Passage if any (liftable conveyor etc).

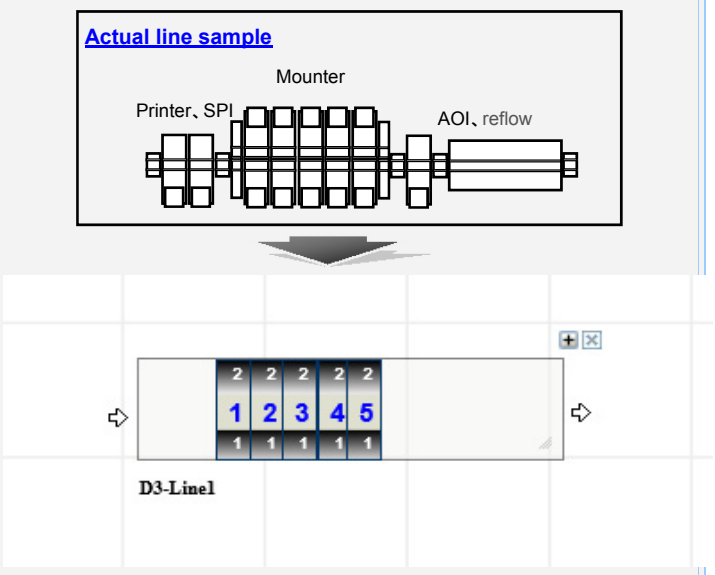
9
After select all, click [Save].



10
Move the location of the target route (line) in drag-and-drop, to determine.
In addition, it is possible to change the size of the target route (line) and drag the right edge of the target route (line), you can change the location of the machine in the target route (line) and drag the machine.

* Grid lines of the screen, 1 trout will be 2 X 2 m. A guideline grid lines, please decide the approximate size and arrangement of the target route (line).

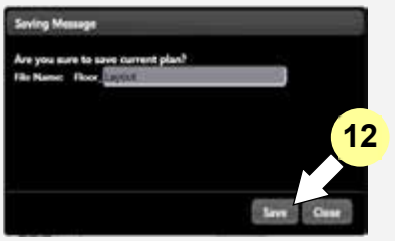
* Printer and reflow such as size and change other than the mounter of the target route (line), please decide the size, including the facility to become an obstacle in the movement.



11
Click  to save the configuration.



12
Saving message appears. Click [Save].



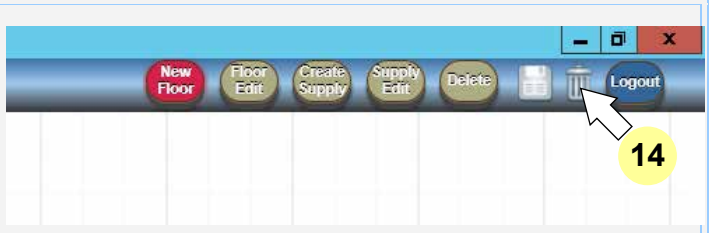
13
After saved successful, click [Close].



14

To finish editing, click  . With this operation, the screen is cleared..

* Before you can save the edited and to carry out this operation Data will be deleted.

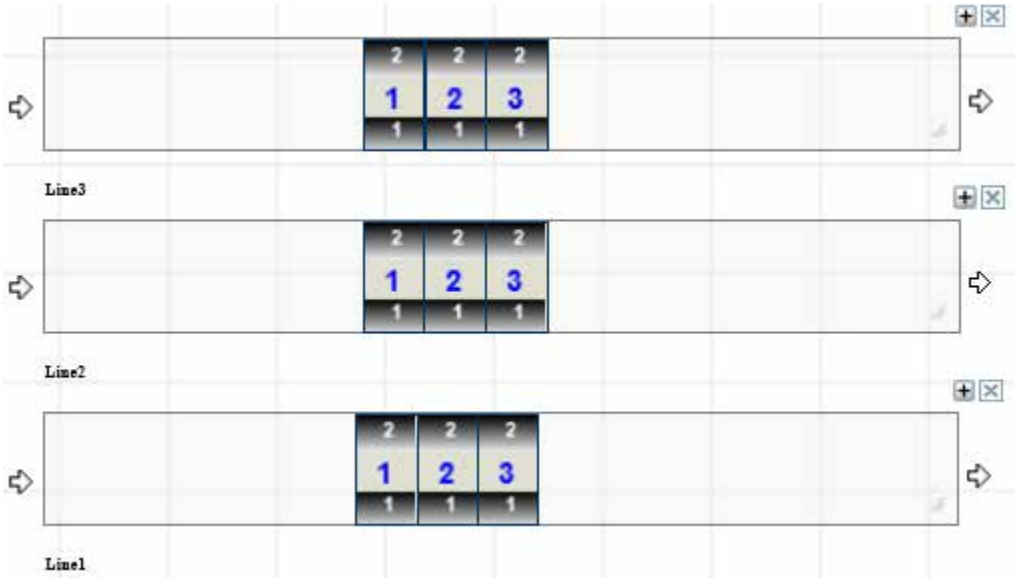


15

After finished configuration, logout from Factory Layout Designer.

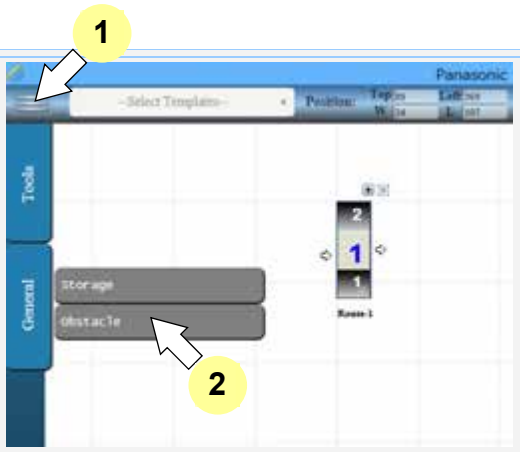


【Note】In the case, The floor layout image of three lines.



2.2.3 Obstacle Layout

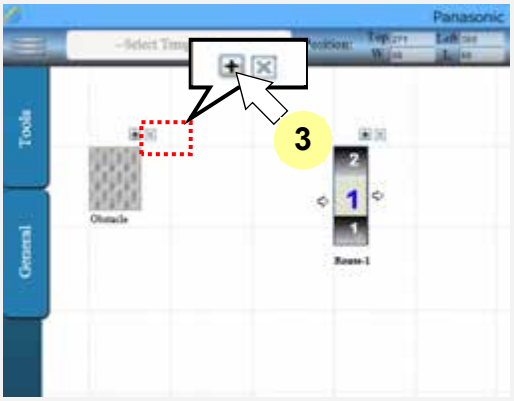
1
Go to Factory Layout Designer and click  to open the side menu.



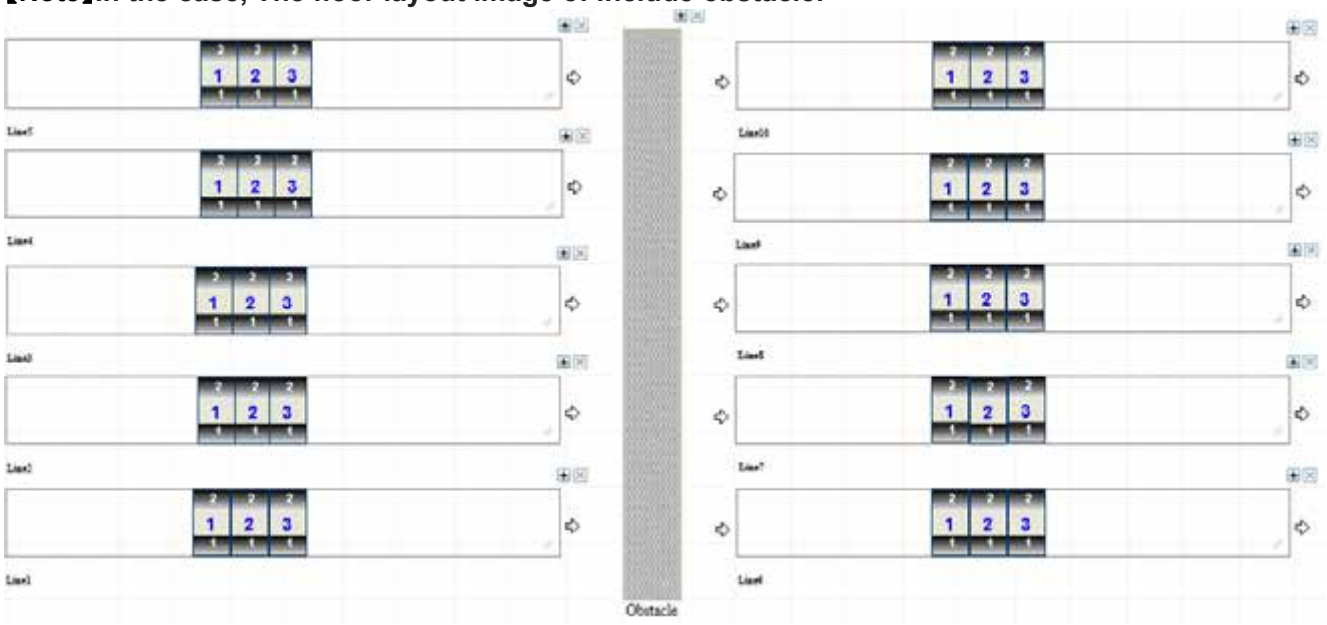
2
Select [General]—[Obstacle].

3
Click [+] at the upper right of an obstacle to see the properties. Drag an obstacle to resize.

After configuration finished, save it in the same way as 2.1.2.



【Note】In the case, The floor layout image of include obstacle.



2.2.4 Configure Supply Area

Configure supply areas for each operator.
The area has two kinds; [Main] and [Support]. [Main] is the area an operator normally works for and [Support] means areas selected in addition to the main area.
See 2.2 for more details.

1
Select [Create Supply] from Factory Layout Designer menu.



2
Select [●] at the front and rear of a machine as supply area.



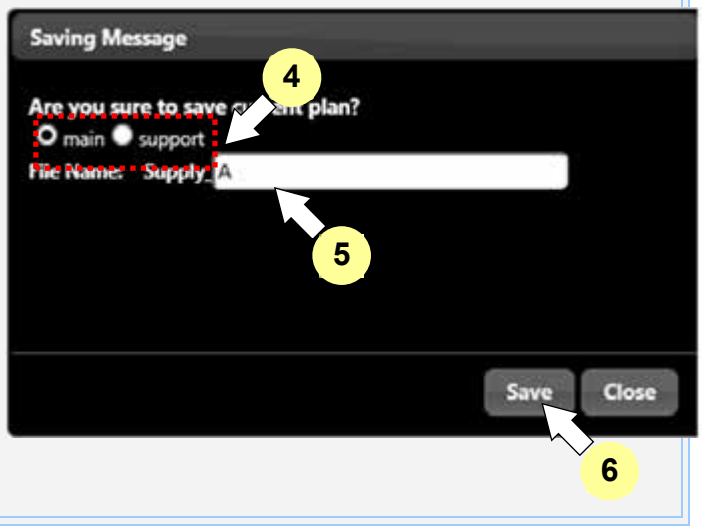
3
Click  to save the configurations.



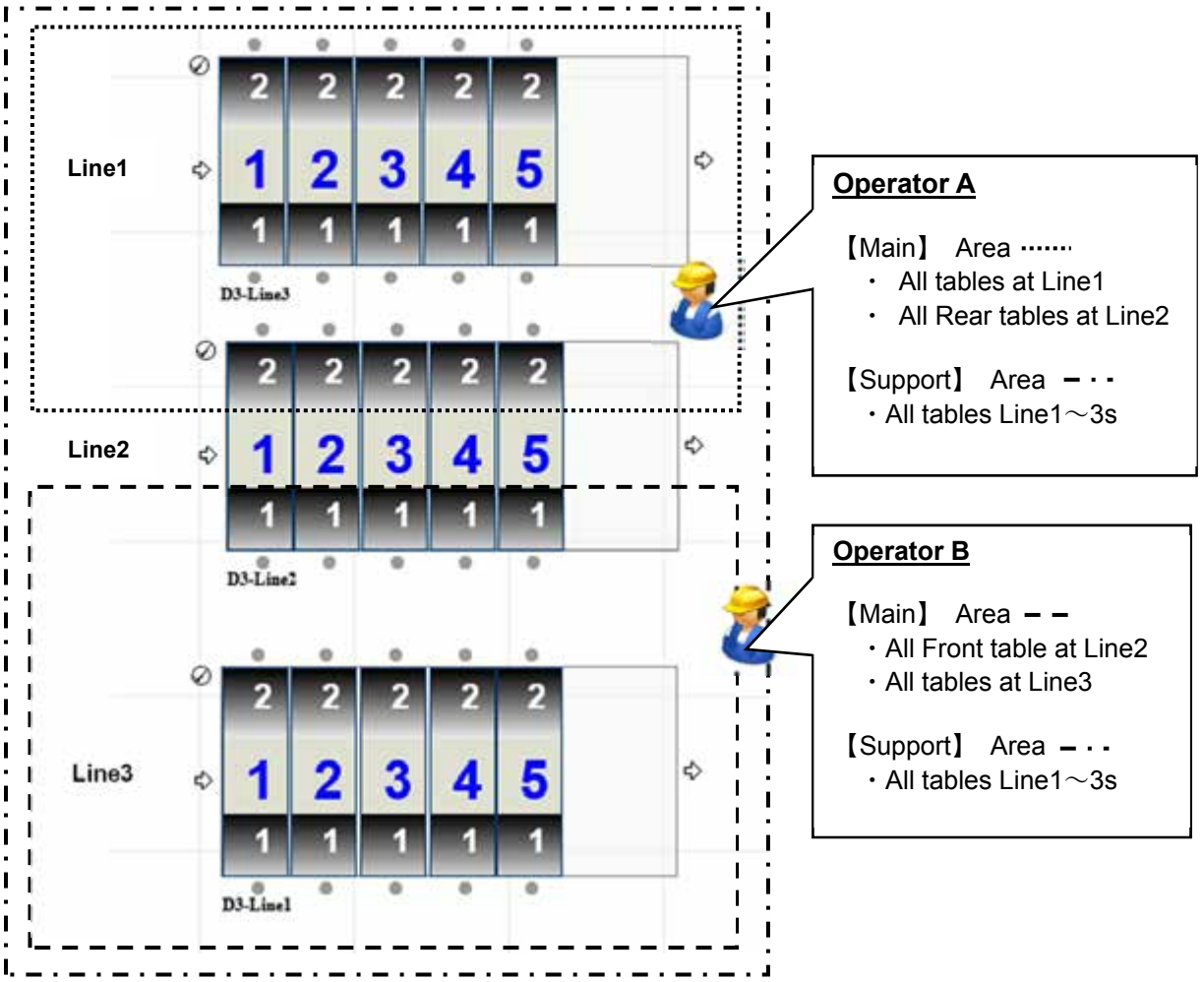
4
Select supply area type (Main or Support).

5
Enter the name.

6
Click [Save].

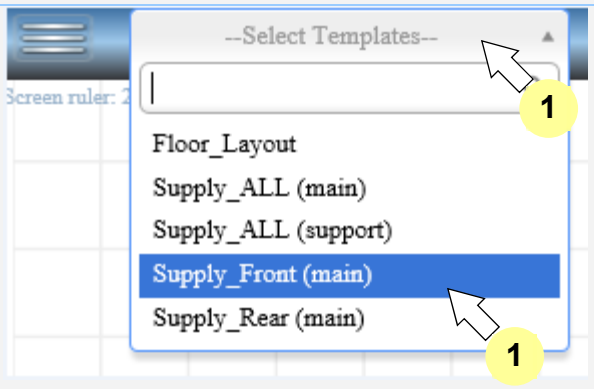


【Note】Two operators supply three lines

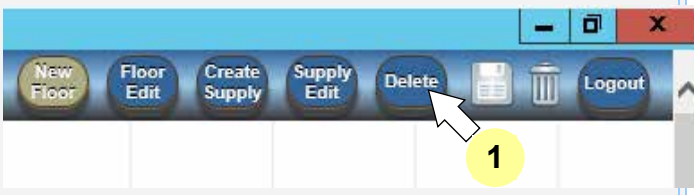


2.2.5 Logout from Factory Layout Designer

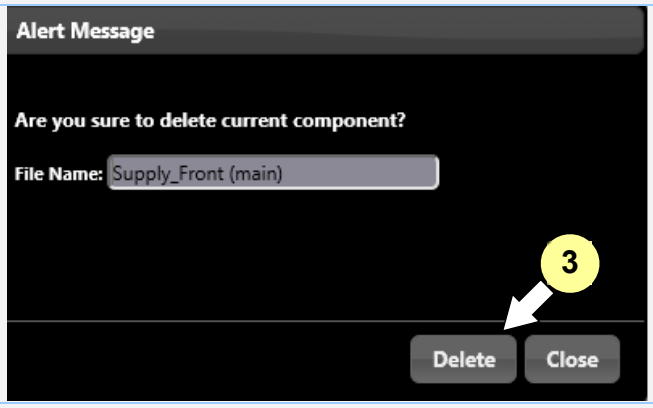
1
Select the Delete object by clicking on the [template selection] in factory layout designer menu.



2
Click [Delete] at Factory Layout Designer menu.



3
Since the alert message is displayed, click the "Delete".



2.2.6 Logout from Factory Layout Designer

1
Click [Logout] at Factory Layout Designer menu.



2.3 Operator Attributes

The following attributes can be set for each operator according to the operators skills.

- Main supply area name
- Support supply area name
- Supply kind
- Supply time: Parts supply (Feeder)
- Supply time: Parts supply (Tray)

Main Supply Area Name

Select a supply area name where the operator usually works for.
See " 2.2.4 Configure Supply Area" for more details.

Support Supply Area Name

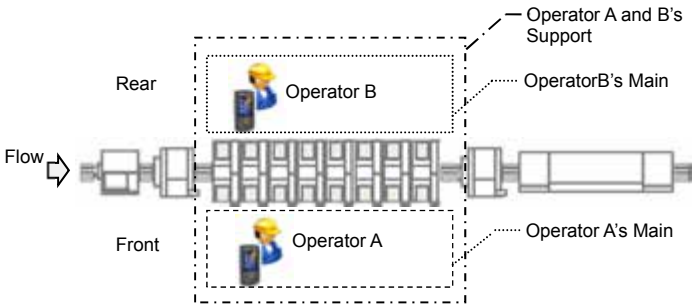
Select an area name assigned additional to the main area.
For example, when the main operator is not in the area, the order will be sent to the support operator.

1. Preferred example

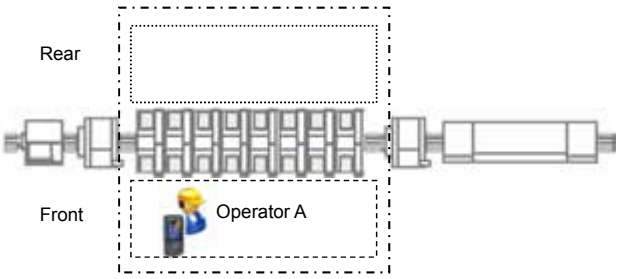
The travel order is explained here when parts supplied by two operators for a line.

■ OperatorA
Main supply area: LineFront
Support supply area: LineFront and Rear

■ OperatorB
Main supply area: LineRear
Support supply area: LineFront and Rear

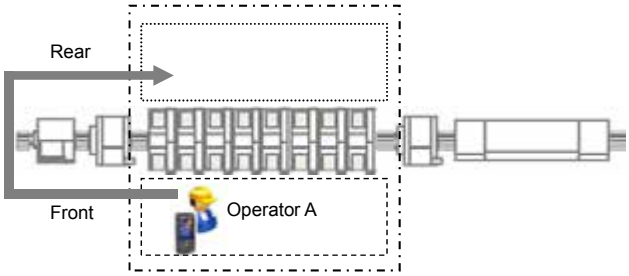


OperatorB logged out from mobile scanner, and no main operator is in LineRear.



Travel order to LineRear will be sent to the support Operator A.

Conversely, log out operator A is from a mobile scanner, usually when the supply personnel has become absent in the front line, you will be to the operator B, travel order to line the front side is issued.

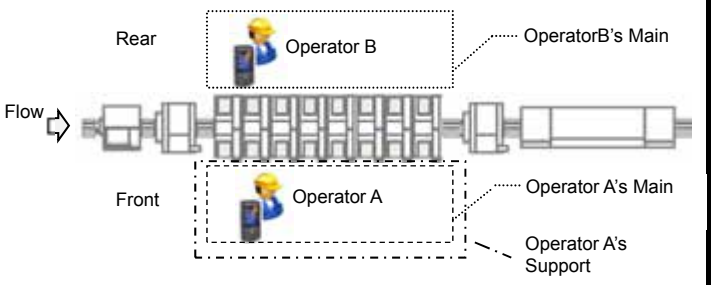


2. Not preferred example

This section describes the undesirable setting example.

■ OperatorA
Main supply area : LineFront
Support supply area: LineFront

■ OperatorB
Main supply area: LineRear
Support supply area: None



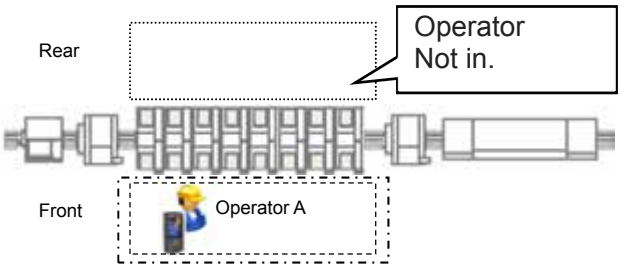
In this way, when the scope of supply in charge operator has not been set is present, the situation such as the following will occur.

Logout of the operatorB is mobile scanner, usually supply the person in charge is absent from the line after the side.

Since the line rear supply charge operator is not set, the travel order to line rear side are not displayed.

It becomes a situation in which the parts supply stagnates.

In order to prevent a situation as described above, with respect to all of the range, please set the supply charge operator.



Supply Kind

Configure the supply area (Feeder/Tray) the operator is in charge.

Supply time: Parts supply (Feeder)

Configure the time needed for the operator to supply (splicing) tape parts.
Configure the time needed to complete supplying and verifying the parts after they are arrived in the machine.
Range: 1～9,999 [sec.]

* If the early person and slow person supply time had been working on the same scope of supply, Travel order is assigned more than slow person feed time in early person supply time.

Supply time: Parts supply (Tray)

Configure the time needed for the operator to supply tray parts.
Configure the time needed to complete supplying and verifying the parts after they are arrived in the machine.
Range: 1～9,999 [sec.]

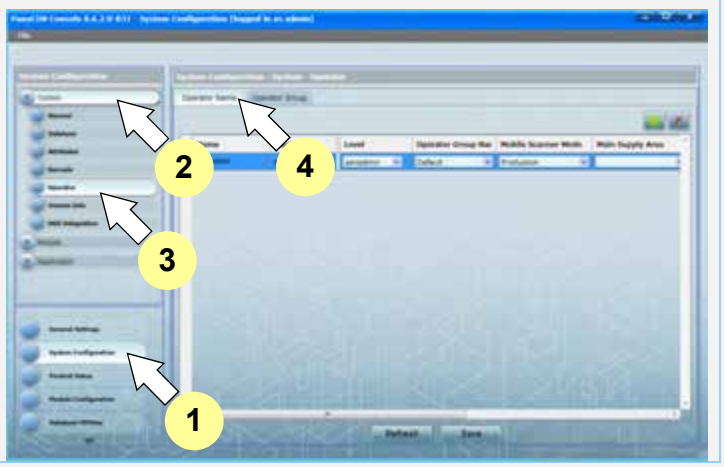
* If the early person and slow person supply time had been working on the same scope of supply, Travel order is assigned more than slow person feed time in early person supply time.

1
Login to PanaCIM Console. Click [System Configuration].

2
Click [System].

3
Click [Operator].

4
Click [Operator] tab.

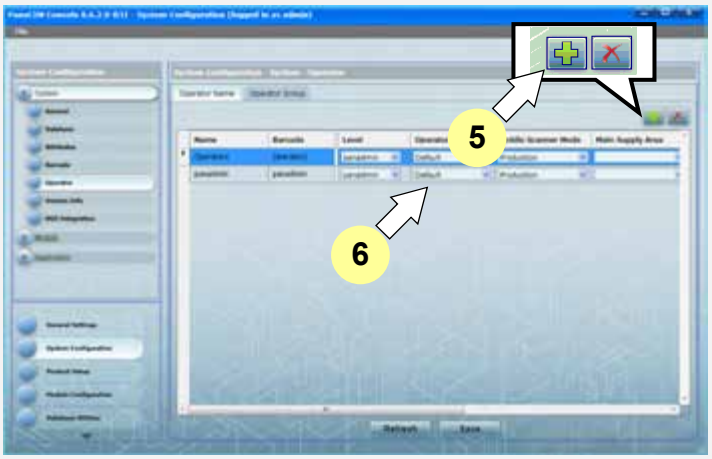


5
Click [+] button to add an operator.

6
Set the "operator group name" for each operator.

* Please set the "operator group name" the privilege of the "travel order" has been set. How to set privilege, please refer to the "About the privilege setting of [Supplement] operator group".

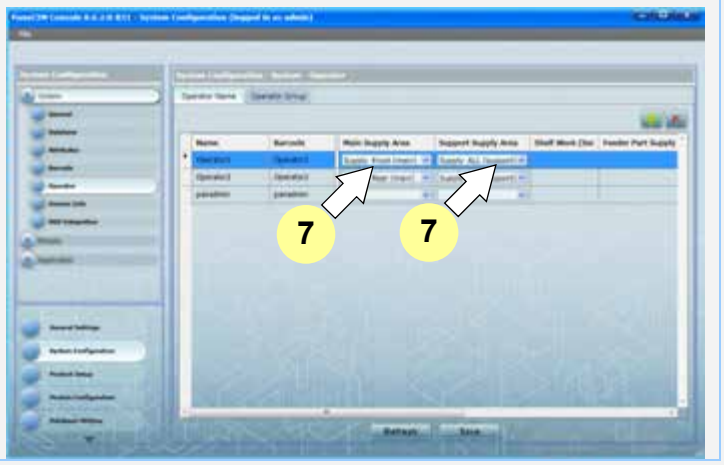
* If the "operator group name" is "Production", it comes with a check in the "travel order". Please remove the check if you do not want to use the component supply navigator.



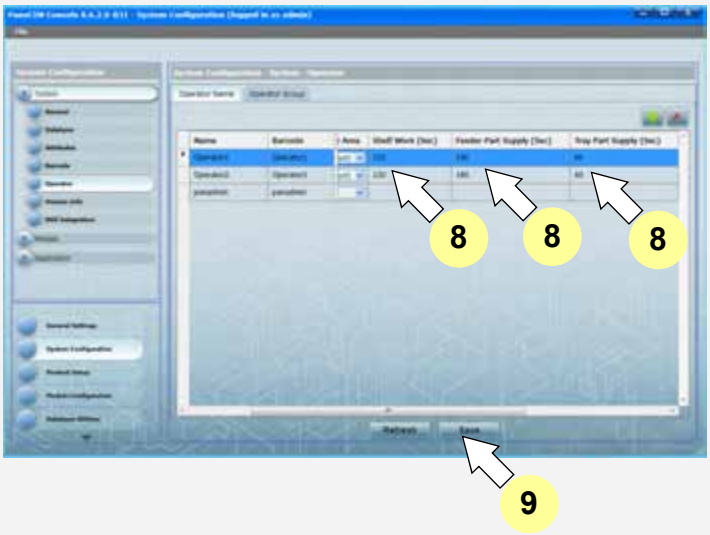
7
Set [Main] and [Support] for each operator.

* At the operator of each production shift, please set to be able to cover the whole line. If a particular line is missing the set as "Main supply area" to "Support supply area", without displaying the travel order of the line, it may interfere with the production.

* The operator has not set a "Main supply area" and "Support supply area ", The travel order of all lines are assigned.

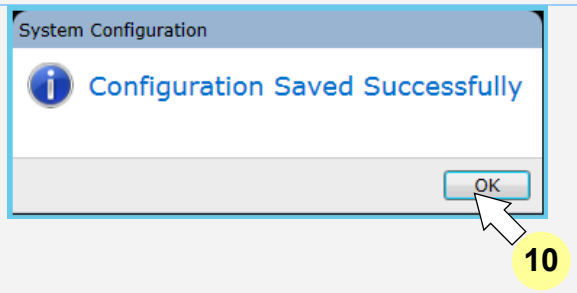


8
Configure (default) times for operators.
- Shelf work (picking from shelf)
- Supply time: Parts supply (Feeder)
- Supply time: Parts supply (Tray)



9
After configuration finished, click [Save].

10
[Configuration Saved Successfully] message appears. Click [OK].



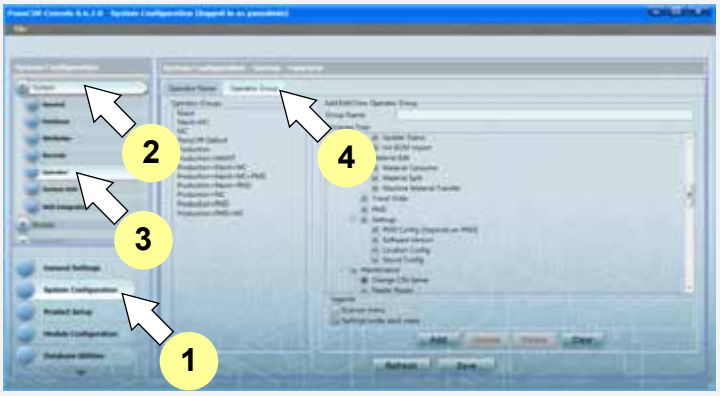
[Supplement] for privileged setting of the operator group
In order to use the parts supply navigation, you need to enable the travel order of the operator group.
Setting procedure is as follows.

1
Login to PanaCIM Console. Click [System Configuration].

2
Click [System].

3
Click [Operator].

4
Click [Operator] tab.

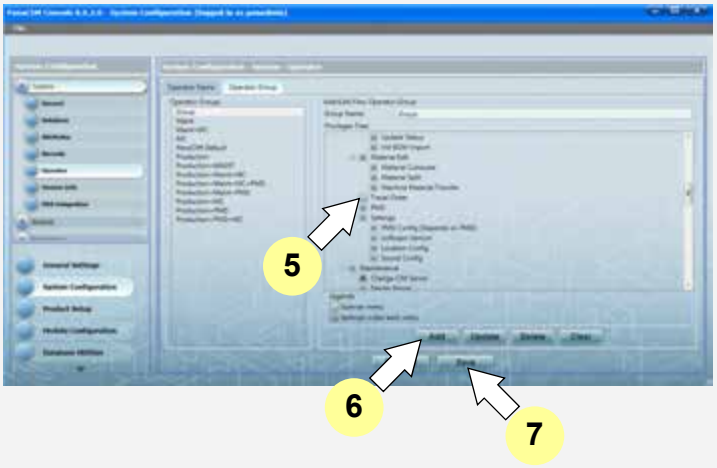


5
Check the "travel order" of privilege tree.

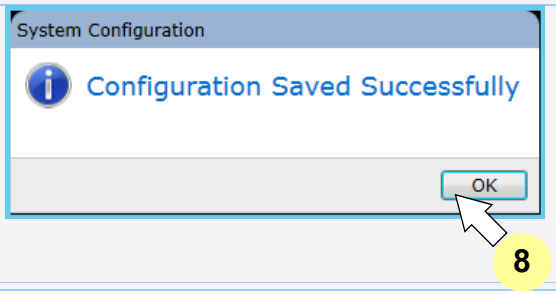
* Please set according to the customer's operation for other privileges.

6
After configuration finished, click [add].

7
Click [Save].



8
[Configuration Saved Successfully] message appears. Click [OK].



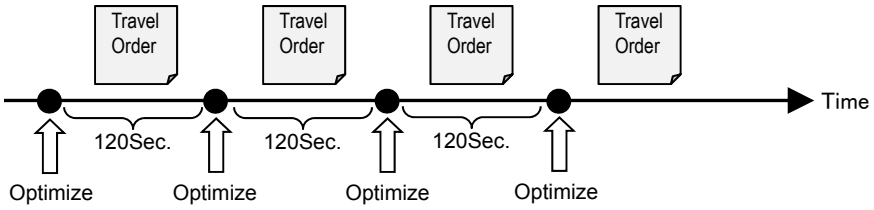
2.4 Travel Order Conditions

Configure the following travel order output conditions.

- Route optimization interval
- Route optimization threshold
- Fixed travel order count
- Part exhaust time priority
- Enable tray supply order
- Supply time: Parts supply (Feeder)
- Supply time: Parts supply (Tray)

Route Optimization Interval

Optimize at the set interval, and create a travel order for each operator.
When this is set to 120 sec, optimize at every 120 sec and create the order.
Range: 60～600 [sec.]



Route optimization threshold

Collect part exhaust data within the set period of time, and create travel orders. When this is set to 180 min, create an order will be created for the data with 180min or less splicing remaining time.

The default value is 180 [min.].
Because recommended not value, change depending on the customer's operational environment.

Range: 0～3600 [min]

* In an option setting of the machine, when the machine operates with "remaining number of warning time of component splicing" turned ON, adjust the warning start time of both the set value and LWS. (If the set value is greater than the warning start time, even if a travel order is displayed, an error occurs after parts verification is carried out.)
  In CM/DT line, "remaining number of warning time of component splicing" cannot be set.

Splicing remaining time	Remaining reel	Machine No.	Feeder	Part name
00:04:50	1714 mm	2	7-L	1005C-08
00:06:25	1882 mm	2	1-L	1005C-09
00:08:40	2382 mm	1	3-L	1005C-02
00:09:27	2462 mm	1	5-L	1005C-03
00:15:13	2814 mm	2	5-L	1005C-07
00:20:03	2002 mm	3	1-L	1005C-13
00:23:38	2066 mm	3	7-L	1005C-16
00:31:07	4502 mm	2	5-L	1005C-11
00:55:04	7042 mm	2	7-L	1005C-12
00:57:21	9022 mm	1	1-L	1005C-01
01:10:55	8722 mm	2	1-L	1005C-05
01:16:43	11662 mn	1	5-L	1005C-03
01:24:54	12520 mn	1	7-L	1005C-04
01:28:37	4446 mm	3	5-L	1005C-15
01:34:21	13780 mn	1	3-L	1005C-02
04:44:32	40002 mn	1	7-L	1005C-04
04:51:00	40002 mn	1	1-L	1005C-01
06:05:50	40002 mn	2	3-L	1005C-06
06:05:50	40002 mn	2	3-L	1005C-10
16:10:31	40002 mn	3	3-L	1005C-14

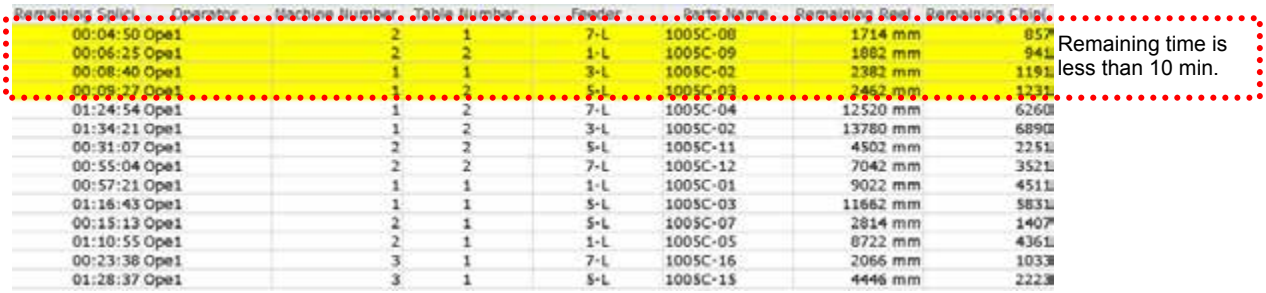
Order not created for data with the splicing remaining time longer than the set time (180min in this case).

Fixed Travel Order Count

Fix the orders with higher priority first. When this is set to 2, the first two orders will not be changed.
Range: 0-10.

Part Exhaust Work Time Priority

When splicing remaining time becomes less than the set time, the operator will be ordered to supply the parts without considering the route.
When this is set to 10 min., the travel order is created with the least splicing remaining time.
If you set the shorter time (such as 1 minute, etc.) you may fail to supply the part even if you make it a high priority. Therefore, you need to consider whether you set a priority to the route or the time for supplying the exhausted part according to your operation environment.
The default value is 600 [s].
Because this is not a recommended value, change it according to your operation environment.
Range: 0 - 3,600[sec].



Remaining Splici...	Operator	Machine Number	Table Number	Feeder	Parts Name	Remaining Reel	Remaining Chip(
00:04:50	Ope1	2	1	7-L	1005C-08	1714 mm	857
00:06:25	Ope1	2	2	1-L	1005C-09	1882 mm	941
00:08:40	Ope1	1	1	3-L	1005C-02	2382 mm	1191
00:09:27	Ope1	1	2	5-L	1005C-03	2462 mm	1231
01:24:54	Ope1	1	2	7-L	1005C-04	12520 mm	6260
01:34:21	Ope1	1	2	3-L	1005C-02	13780 mm	6890
00:31:07	Ope1	2	2	5-L	1005C-11	4502 mm	2251
00:55:04	Ope1	2	2	7-L	1005C-12	7042 mm	3521
00:57:21	Ope1	1	1	1-L	1005C-01	9022 mm	4511
01:16:43	Ope1	1	1	5-L	1005C-03	11662 mm	5831
00:15:13	Ope1	2	1	5-L	1005C-07	2814 mm	1407
01:10:55	Ope1	2	1	1-L	1005C-05	8722 mm	4361
00:23:38	Ope1	3	1	7-L	1005C-16	2066 mm	1033
01:28:37	Ope1	3	1	5-L	1005C-15	4446 mm	2223

Enable Tray Supply Order

Check here to include tray supply in travel orders.

Supply Time: Parts Supply (Feeder)

Set default time needed for tape reel parts supply work (splicing etc.)
This can be set for each operator (see 2.3.1), but this value will be applied when unregistered operator logged in.
Range: 1 - 9,999[sec.].

Supply Time: Parts Supply (Tray)

Set default time needed for tray parts supply work.
This can be set for each operator (see 2.3.1), but this value will be applied when unregistered operator logged in.
Range: 1 - 9,999[sec.].

Configure as follows.

1
Login to PanaCIM Console. Click [System Configuration].

2
Click [Module].

3
Click [Production Monitor and Dispatch].

4
Click [Component supply Optimizer] tab.

5
After configuration finished, click [Save].

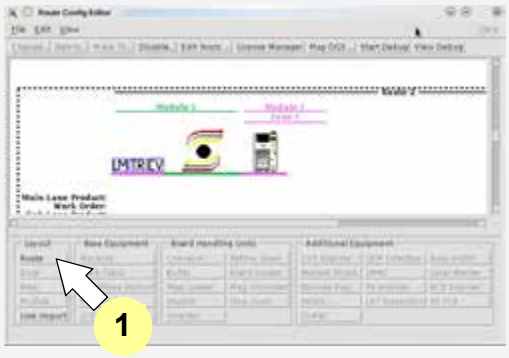


2.5 In-line Part Shelf

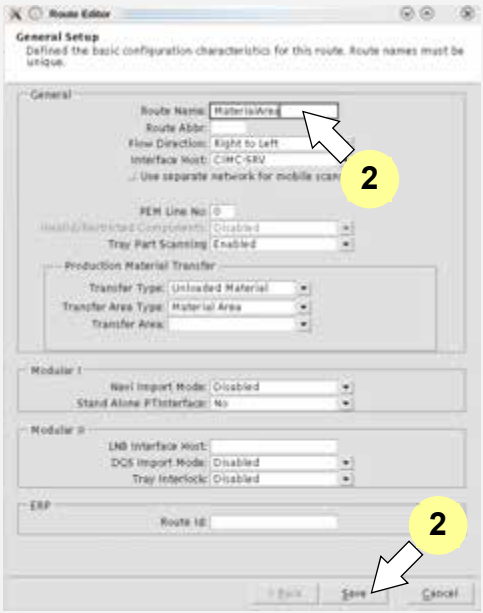
This configuration will be available only with valid [Material Management (Basic)] and [Material Management (Reservation)] License.

2.5.1 Register In-line Part Shelf

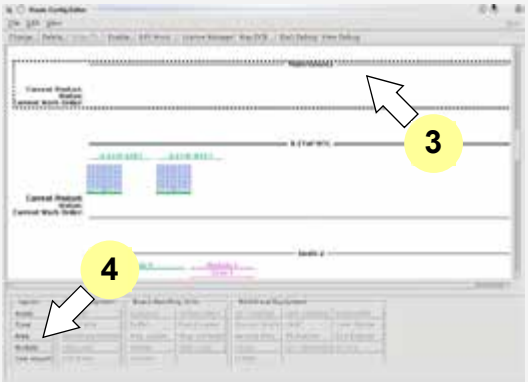
1
Open Route Editor and click [Layout]-[Route].



2
Enter a route name and click [Save].

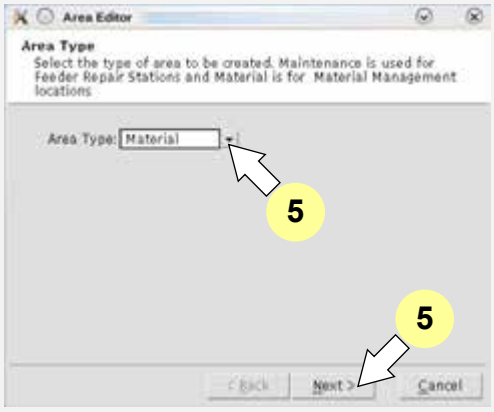


3
Go back to Route Editor and select the route created on Layout screen “2”.



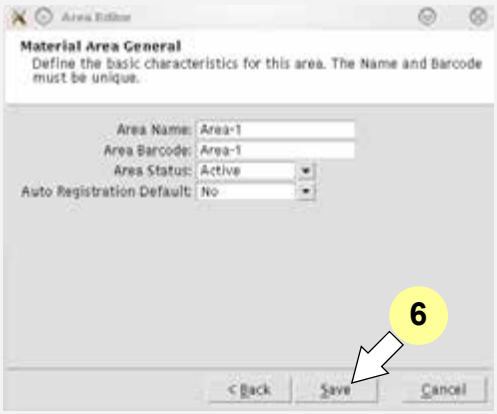
4
Click [Layout] and [Area].

5
Select Area Type: [Material] and click [Next>].

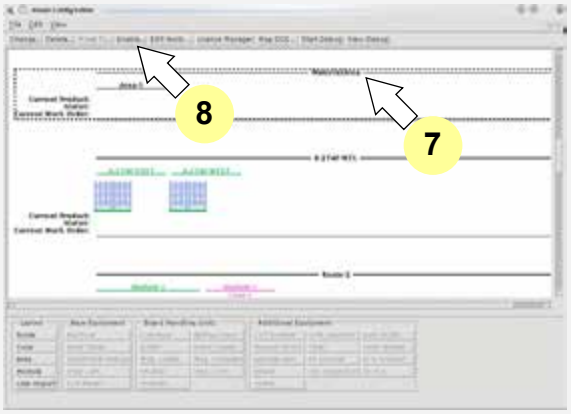


6
Select area information and click [Save].

* in connection with the material area that was created line here, and set it as the line side parts shelf.
(☞ in-line Part shelf)

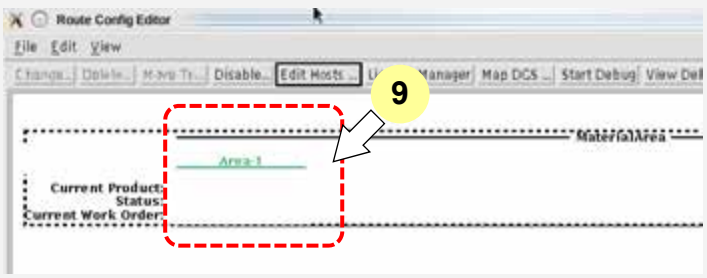


7
Go back to Route Editor and select a route on Layout screen. Click [Enable] to enable the route.



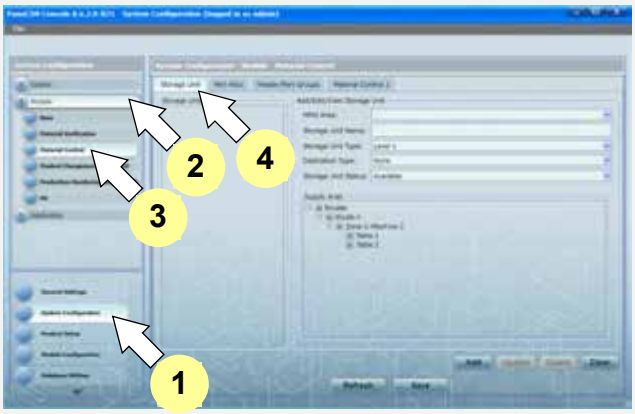
8
Click [Enable] to enable the route.

9
After the route is enabled, its area turns to green.



2.5.2 Configure In-line Part Shelf

1
Login to PanaCIM Console. Click [System Configuration].



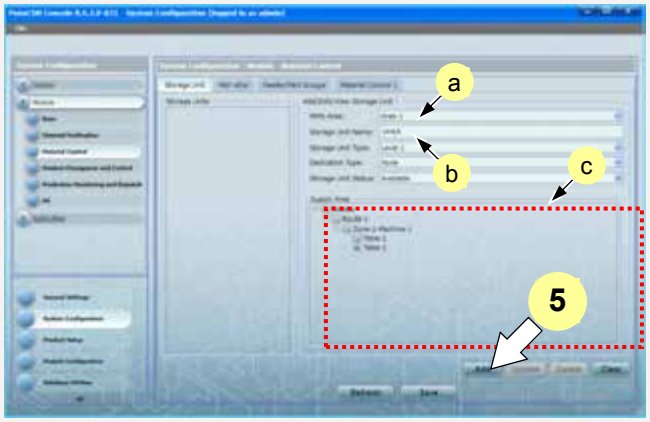
2
Click [Module].

3
Click [Material Control].

4
Click [Storage Unit] tab.

5
Set as follows and click [Add].

a	MMS Area	Select a part shelf
b	Storage Unit Name	Enter part shelf name
c	Supply Area	Select a machine table to supply parts using this storage unit



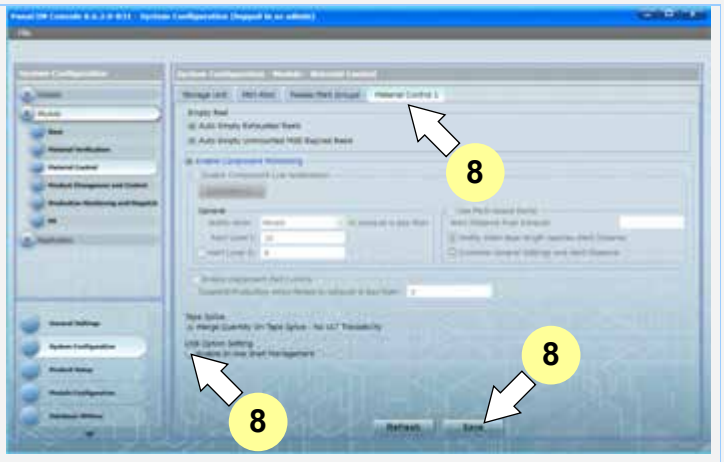
6
After the part shelf was added at [Storage Unit], click [Save].



7
[Configuration Saved Successfully] appears. Click [OK].



8
Click [Material Control] tab, check [Enable In-line Shelf Management] and click [Save].

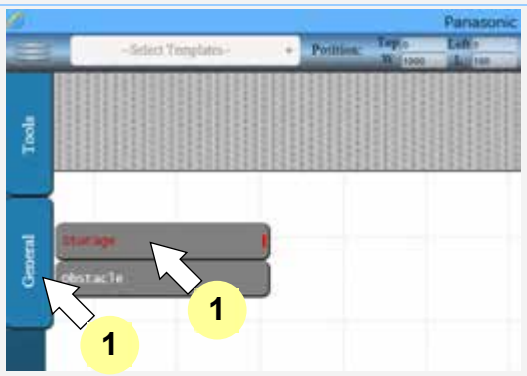


9
[Configuration Saved Successfully] appears. Click [OK].

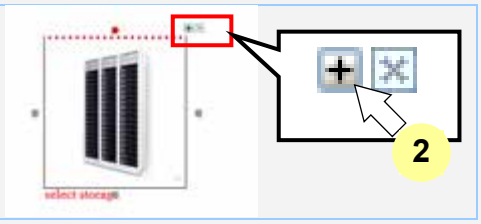


2.5.3 In-line Part Shelf Layout

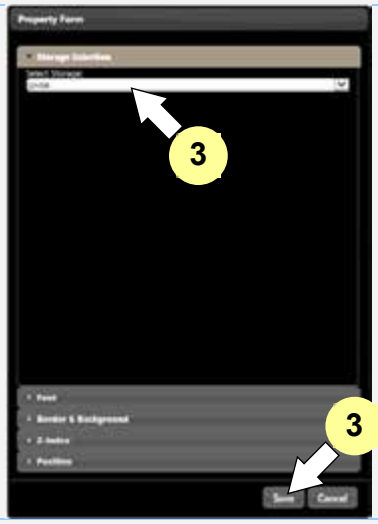
1
Login to Factory Layout Designer, and select [General - [Storage Unit].



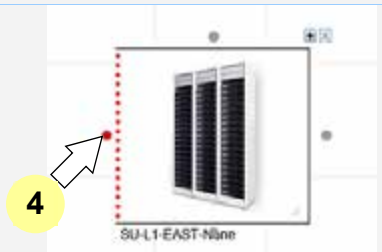
2
Click [+] at upper right of the added storage unit image.



3
Select a part shelf as storage and click [Save].



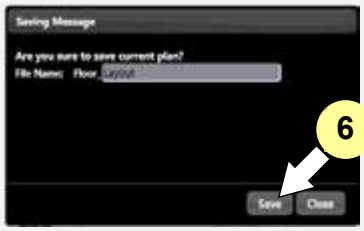
4
Click [●] at the sides of the image where operator can pick parts from.



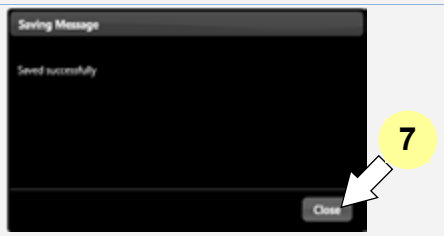
5
Go back to Factory Layout Designer screen, and click  to save the configuration.



6
[Saving Message: Are you sure to save current plan?] appears and click [Save].



7
[Saved successfully] message appears. Click [Close].

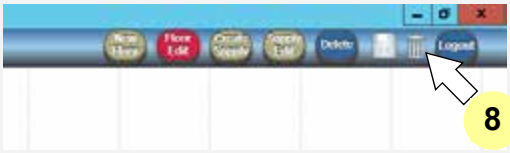


8

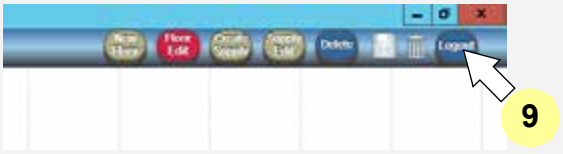
When you move on to the next editing, click  .

With this operation, the screen is cleared.

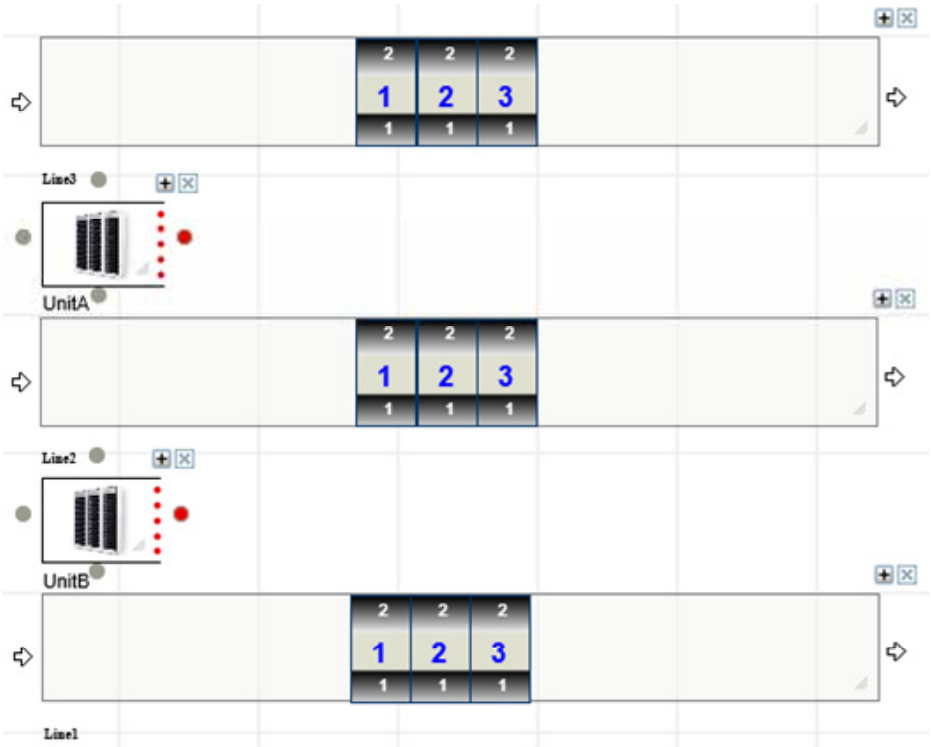
* Before you can save the edited and to carry out this operation Data will be deleted.



9
After finished configuration, logout from Factory Layout Designer.



【Note】In the case, The floor layout image of include in-line shelf.



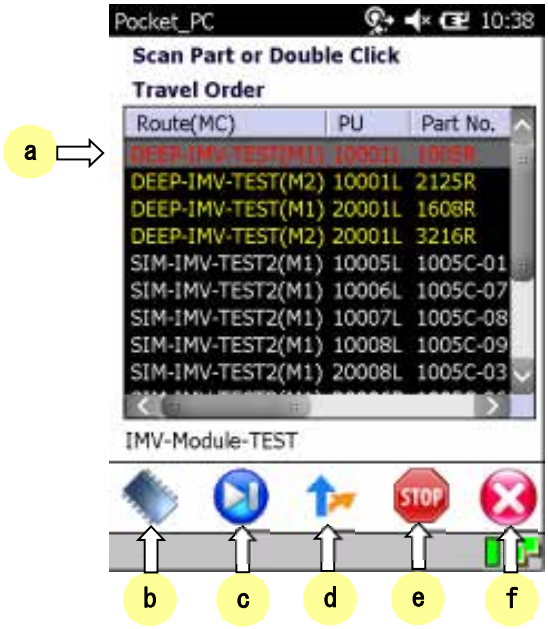
3. Travel Order

3.1 Screen Configuration

The verification system has the following two modes; Non-consolidated edition and Consolidated edition (*1).
The part scan menu of these two modes is different from each other.

*1 The consolidated edition supports only NPM series connected to LNB. The NPMX included line does not support the non-consolidated edition.

3.1.1 Verification Mode in Consolidated Edition / Non-consolidated Edition (Import of Initial Quantity: No)



No.	Item	Details
a	Travel Order	Shows the address that needs part supply. They are shown in the order of priority with the following information. Route(MC) Route name and (machine number). PU Part address For each digit, it becomes the following content. X***** : Table number *YYYY* : Feeder address *****Z : Feeder sub address Feeder address will be the following. In the case of tape feeder : 0001-0030 In the case of single tray or twin tray on the left side : 1001-1020 In the case of twin tray on the right side : 2001-2020 Feeder sub address is the following. Feeder left (in the case of a double feeder / thin type tape feeder), or in the case of a single feeder: “L”

		Feeder right (in the case of a double feeder / thin type tape feeder), or in the case of a single feeder: “R” In the case of tray: empty Part Number: Part number set in production program SU: Storage unit name *(SU stands for Storage Unit) Time to Splice: Remaining time until splicing limit length *The default splicing length limit value is 1,200mm, and for LNB, it is alterable with LWS (in CM/DT line, it is not alterable). *The calculation formula is “The number of remaining chips until the splicing limit length” * “Cycle time” / “Vacuum count per sheet” ‘Cycle time’ and ‘Vacuum count’ are calculated from the data of actual production. Remaining Board Count: The number of boards that can be produced before part exhaust. Remaining Reel Length: Remaining length of reel The color of characters changes according to the status of parts. White: OK *Other than below (warning, emergency warning, parts exhaust), parts corresponding to the setting of route optimization threshold. Yellow: Warning *Parts corresponding to the "Remaining setting for warning" of LWS. For the setting method, refer to the instruction manual of the LNB. *For CM/DT line, it is applicable only to feeders, and can be altered with the settings for the remaining warning setting tool (CRPT) installed to PT200. For info on how to change the settings, refer to “■How to change the settings for the remaining warning setting tool (CRPT).” Orange: Emergency Warning * Parts corresponding to the "Remaining count urgent warning" of LWS. For the setting method, refer to the instruction manual of the LNB. (In CM/DT line, the urgent warning message is not displayed.) Red: Part Exhaust
b	Show Part Verification Menu	Go to Part Verification Menu screen.
c	Order Skip	When ordered parts cannot be supplied, skip the parts.
d	Reroute	Run route optimization forcibly and update the order. Select from two options; only the operator who selected this or all operators.
e	Stop Receiving Orders	Stop receiving travel orders for the selected route. Use this to hide orders for a route stopped for some reason. Orders appear again automatically when the route starts production (boards start moving).
f	Close	Close the screen.

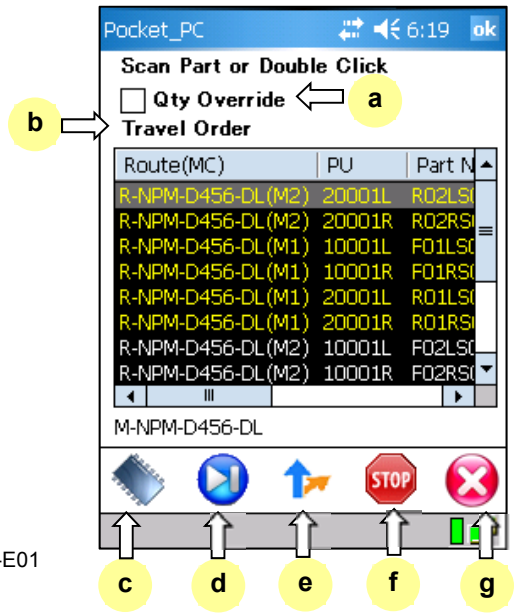
■ How to change the settings for the remaining warning setting tool (CRPT)

You can configure the remaining warning settings by changing each item for [WarningInfomaition] described in the setting file, “C:¥CRPT¥CRPT.ini,” for the remaining warning setting tool (CRPT) installed on the PC of PT 200.

[WarningInfomaition]	Setting item
WarningLevel1Use=1	Warning use 1: Yes, 0: No
WarningStartTimeFeeder1=600	Feeder warning start time (Unit: second) When the time it takes for a reel installed to feeder to reach its splicing length limit falls below the set time limit, the remaining warning will be started.
WarningStartProdSheet1=10	Warning start possible production volume (Unit: piece) When the maximum number of pieces that can be produced by the machine falls below the set volume limit, the remaining warning will be started.
WarningStartLengthFeeder1=2000	Feeder warning start length (Unit: mm) When the remaining length of tape installed to feeder is equal to (the set length + a splicing length limit of 1,200mm), the remaining warning will be started.

3.1.2 Verification Mode in Non-consolidated Edition (Import of Initial Quantity: Yes)

*It is effective only when verification mode on PanaCIM is MV.



3-1001-E01

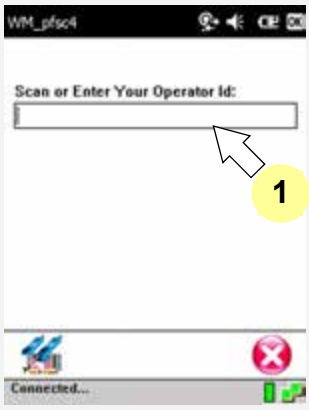
No.	Item	Details
a	Qty Override	This is displayed when “Automatically import quantity during verification” is enabled. Marking the checkbox of Qty Override allows you to enter the quantity on the verification screen.
b	Travel Order	Shows the address that needs part supply. They are shown in the order of priority with the following information. Route(MC) Route name and (machine number) PU The position of the feeder installing parts is displayed Details of each digit are as follows: X***** : Table number *YYYY* : Feeder address *****Z :Feeder sub-address Feeder address will be the following. In the case of tape feeder : 0001-0030 In the case of single tray or twin tray on the left side : 1001-1020 In the case of twin tray on the right side : 2001-2020 Feeder sub address is the following. Feeder left (in the case of a double feeder / thin type tape feeder), or in the case of a single feeder: “L” Feeder right (in the case of a double feeder / thin type tape feeder), or in the case of a single feeder: “R” In the case of tray: empty Part Number Part number set in the production program SU Storage unit name *(SU stands for Storage unit)

		Time to Splice: Remaining time until the splicing length limit is reached *The default splicing length limit value is 1,200mm, and for LNB, it is alterable with LWS (in CM/DT line, it is not alterable). *The calculation formula is “The number of remaining chips until the splicing limit length” * “Cycle time” / “Vacuum count per sheet” ‘Cycle time’ and ‘Vacuum count’ are calculated from the data of actual production. Remaining Board Count: The number of boards that can be produced before part exhaust. Remaining Reel Length: Remaining length of reel The color of characters changes according to the status of parts. White: OK *Other than below (warning, emergency warning, parts exhaust), parts corresponding to the setting of route optimization threshold. Yellow: Warning *Parts corresponding to the "Remaining setting for warning" of LWS. For the setting method, refer to the instruction manual of the LNB. * For CM/DT line, it is applicable only to feeders, and can be altered with the settings for the remaining warning setting tool (CRPT) installed to PT200. For info on how to change the settings, refer to “■How to change the settings for the remaining warning setting tool (CRPT) in 3.1.1 Verification Mode in Consolidated Edition / Non-consolidated Edition (Import of Initial Quantity: No).” Orange: Emergency Warning * Parts corresponding to the "Remaining count urgent warning" of LWS. For the setting method, refer to the instruction manual of the LNB. (In CM/DT line, the urgent warning message is not displayed.) Red: Part Exhaust
c	Show Part Verification Menu	Go to Part Verification Menu screen.
d	Order Skip	When ordered parts cannot be supplied, skip the parts.
e	Reroute	Run route optimization forcibly and update the order. Select from two options; only the operator who selected this or all operators.
f	Stop Receiving Orders	Stop receiving travel orders for the selected route. Use this to hide orders for a route stopped for some reason. Orders appear again automatically when the route starts production (boards start moving).
g	Close	Close the screen.

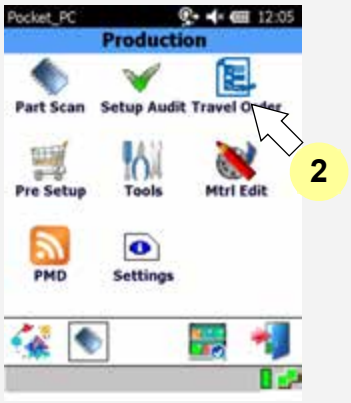
3.2 Show Travel Order

3.2.1 Verification Mode in Consolidated Edition / Non-consolidated Edition

1
Scan or enter operator ID to a mobile scanner.



2
Select [Travel Order] when Menu screen appears.



3
Travel Order screen appears.

* Travel orders that are displayed on the screen are not assigned to other operator to exit the screen. When you interrupt the parts component work, please exit this screen.

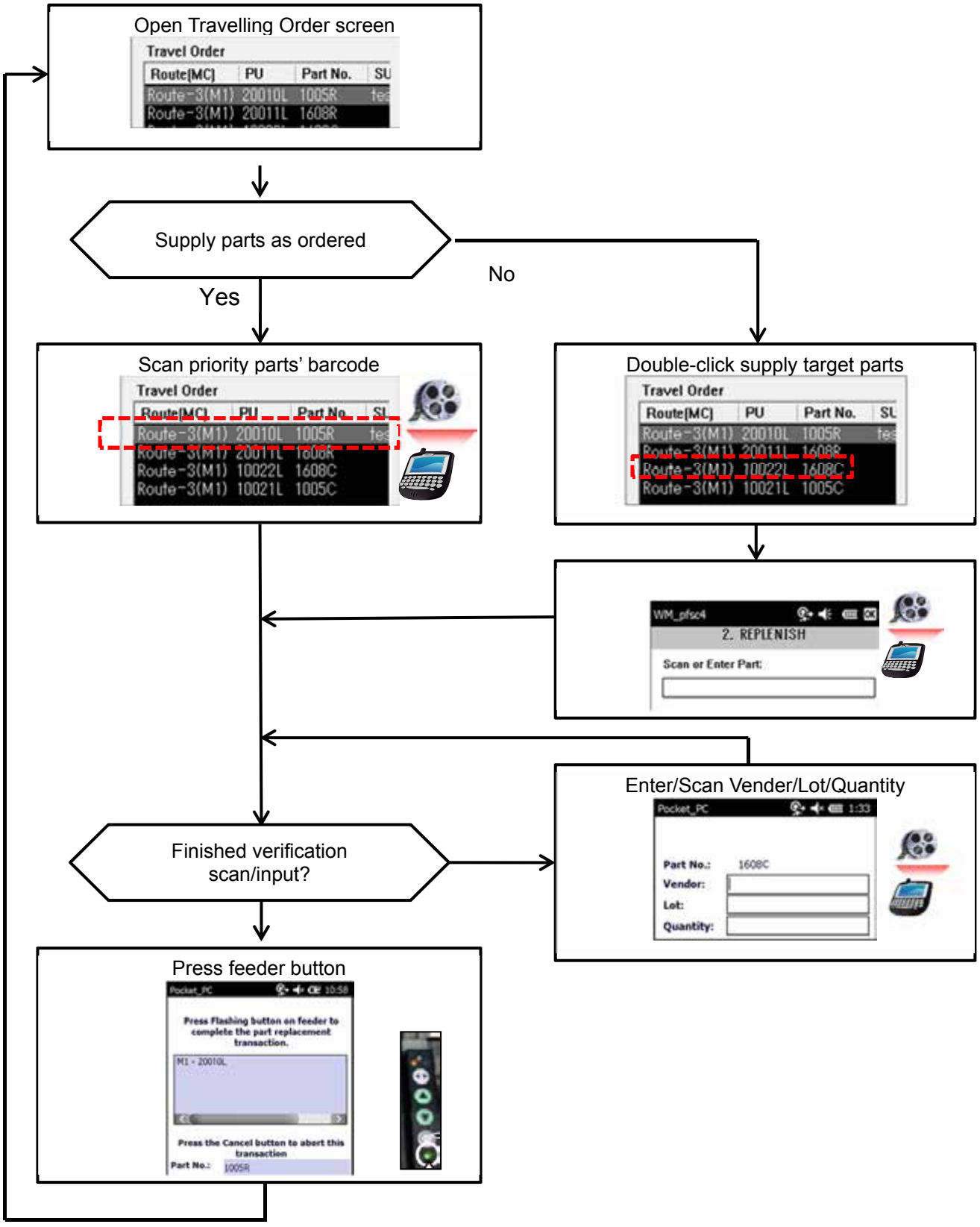
* Screen missing forget as prevention you can use the "Auto Logout" function. This function is, past the time set, is the ability to log out automatically when you are not using the scanner. Setting method, please refer to the instruction manual of the "material verification".

* If the scanner of the battery has expired while working, please log in again from another scanner. If you do not want to re-login, no instruction for the operator who were logged at the time the battery has expired is cleared.



3.3 Part Verification

3.3.1 Work Flow



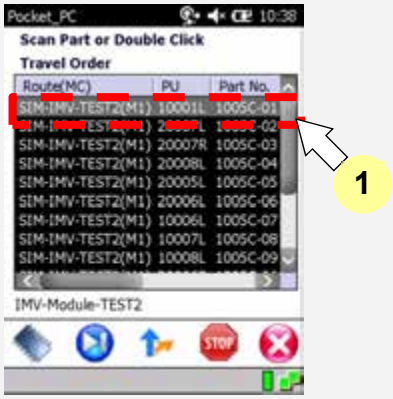
3.3.2 Part Verification From Travel Order Screen

■Verification Mode in Consolidated Edition / Non-consolidated Edition (Import of Initial Quantity: No)

1
Scan part barcode at the top of Travel Order list.

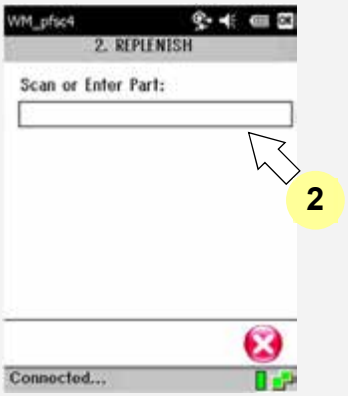
- Go to “4” if you don’t need to add anything.
- Go to “3” to add something.

*To supply parts that are not at the top, follow Step”2” and double-click it on the list.



2
Scan or enter the barcode of the parts that were double-clicked at STEP“1”.

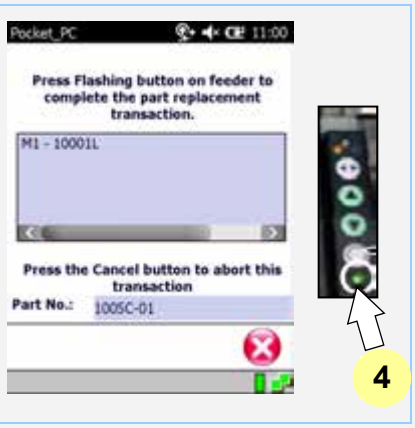
- Go to “4” if you don’t need to add anything.
- Go to “3” to add something



3
Scan or enter the additional parts’ barcode. The n go to Step”4”.



4
Press flashing button on the feeder when its LED lamp turns to green.



5
Go back to Travel Order screen.



【Note】Part verification with warning

When a warning such as remaining part count is detected, the order will turns to yellow.

To supply the parts with warning in preference, scan the parts' barcode or double-click the order. The screen changes to the parts' verification screen.



3.3.3 Part Verification from Travel Order Screen (Override)

■Verification mode in non-consolidated edition (Import of initial quantity: Yes)

*It is effective only when verification mode on PanaCIM is MV.

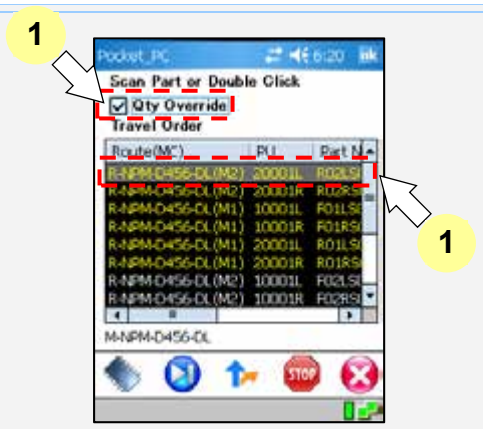
*The **Qty Override** function on the travel order screen does not support consolidated verification mode.

1

Mark the checkbox of Qty Override in Travel Order list and scan the barcode.

- Go to “4” if you don’t need to add anything.
- Go to “3” to add something.

* To supply a part other than the topmost part, double-click the part in the list according to STEP “2”.

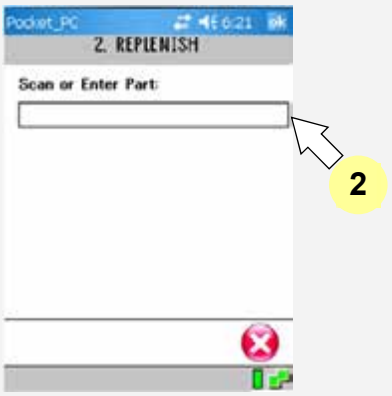


3-3007-J01

2

Scan or enter the barcode of the parts that were double-clicked at STEP “1”.

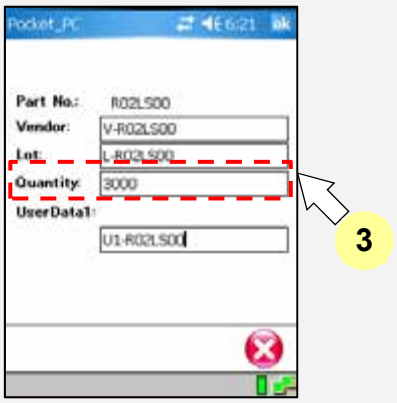
- Go to “4” if you don’t need to add anything.
- Go to “3” to add something.



3-3008-J00

3

The input field of Quantity appears as additional part information input. Input the value.
After entering the value, go to step “4”.



3-3009-J00

4

After the LED lamp on the feeder flashes, press the **confirm** button on the feeder.



3-3010-J01



5

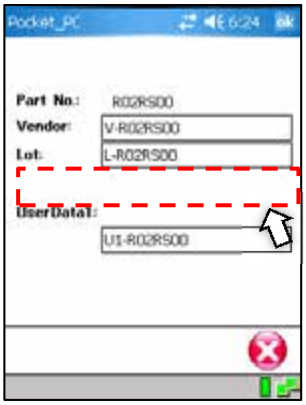
Go back to Travel Order screen.



3-3011-J01

[Note] The input screen of part information in case the checkbox of 'Qty Override' is not marked.

If you do not mark the checkbox of Qty Override, the default quantity is imported. Therefore, the input field of STEP "3" does not appear on the screen.



3-3012-J01

3.3.4 Part Verification during Machine Stop Due To Wrong Supply Work

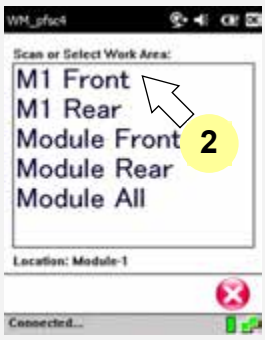
This section explains how to verify parts when machine stopped due to [feeder clog] or [no splicing check].

■Verification mode in consolidated edition

1
Press  on Travel Order screen.



2
Select an error-stopped area.



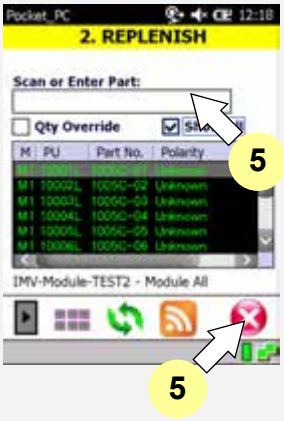
3
Press [Setup] on Setup screen.



4
From the list that appears, select the "RECHECK".



5
Scan the parts barcode or enter the parts name. After the error is cleared, press  and go back to Supply Order screen.

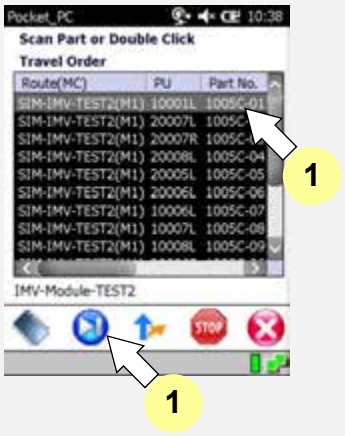


3.4 Skip Order

Skip an order when its supply work cannot be done for some reason.

■Verification mode in consolidated edition / non-consolidated edition

1
Select an order to skip on Travel Order screen and press  .



2
Press  on the confirmation screen.



3
Skipped order will disappear.

* To re-display, back to the menu screen, will be displayed again, the transition to a component supply travel order screen.



3.5 Reroute

If you want to re-run the route optimization for some reason, you can run under the following operation.

■Verification mode in consolidated edition / non-consolidated edition

1
Press  on Travel Order screen.



2
Confirmation screen is displayed.
When the user presses  run the route optimization of all operators, to re-create the travel orders.

Run the route optimization of the only operator that you performed by pressing  to re-create the travel orders.



3
It appears travel order that has been re-created.

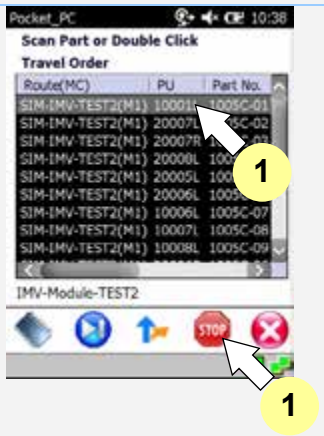


3.6 Stop Receiving Orders

If the components supply for any reason (such as is stopped temporarily production) there are unnecessary route, you can stop receiving temporarily travel order by the following operation.

■Verification mode in consolidated edition / non-consolidated edition

1
In supply travel order screen, select the travel order of route you want to stop the travel orders, and press STOP.



2
Press on the confirmation screen.



3
The specified route travel order receiving stop, will not be displayed. (Travel order other than the specified route will be displayed)

* When the production of the executed route is started (PCB flows out), it will be re-displayed automatically.

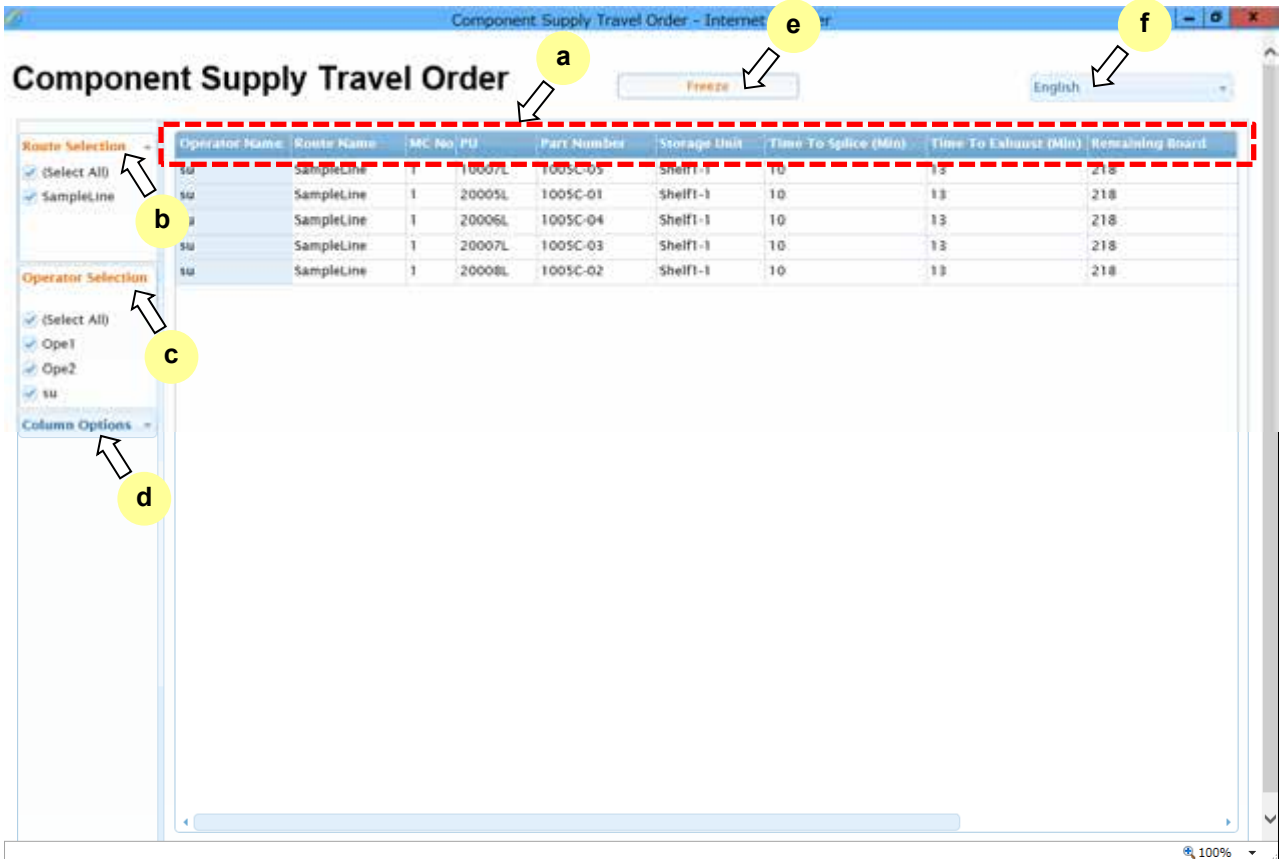


= MEMO =

4. Monitor Supply Work Progress

4.1 Screen Configuration

You can see the components supply travel orders to all operators of the line that you want to manage.
This display will be updated automatically, you can centrally monitor the supply work progress of all operators.



No.	Item	Details
a	Operator Name	The name of the operator who receives the order.
	Route Name	The name of the route travelling to (line name).
	MC No	Machine number (serial number from the line head)
	PU	Feeder address for each digit, it becomes the following content. X***** : Table number *YYYY* : Feeder address *****Z : Feeder sub address Feeder address will be the following. In the case of tape feeder : 0001-0030 In the case of single tray or twin tray on the left side : 1001-1020 In the case of twin tray : 2001-2020 Feeder sub address is the following. Feeder left (in the case of a double feeder / thin type tape feeder), or in the case of a single feeder: “L” Feeder right (in the case of a double feeder / thin type tape feeder), or in the case of a single feeder: “R” In the case of tray: empty
	Part Number	The part name registered on DGS

	Storage Unit	The name of the shelf where the parts are stored
	Time to Splice (Min)	Remaining time until splicing limit length *The default splicing length limit value is 1,200mm, and for LNB, it is alterable with LWS (in CM/DT line, it is not alterable). * A formula: Chip number of remaining until the splicing limit length * cycle time / pickup count of times per sheet ". "Cycle time" and " pickup count" is calculated from the data in the actual production.
	Time to Exhaust (Min)	Time to parts exhaust
	Remaining Board	The number of boards that can be produced before part exhaust
	Remaining Parts	Remaining number of parts being used in the on-going production
	Remaining Reel Length (mm)	Remaining reel length *not applied for trays
b	Route Selection	Shows checked route information only
c	Operator Selection	Shows checked operator information only
d	Column Option	Select column(s) to display
e	Freeze/Update	[Freeze] stops updating the information, and [Update] to start updating
f	Language	Select a language for the display from Japanese, English and Chinese.

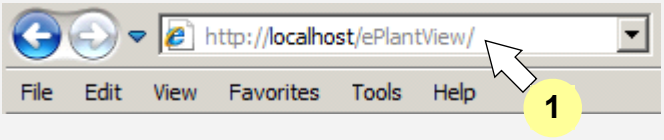
∗For the background color of the components supply patrol instruction
The color of characters changes according to the status of parts.

Background color	Content	Details
White	OK	Below (warning, emergency warning, parts exhaust) other than, parts corresponding to the set of route optimize threshold are eligible.
Yellow	Warning	Parts corresponding to the "Remaining setting for warning" of LWS screen are eligible. For the setting method, refer to the instruction manual of the LNB. For CM/DT line, it is applicable only to feeders, and can be altered with the settings for the remaining warning setting tool (CRPT) installed to PT200. For info on how to change the settings, refer to "■How to change the settings for the remaining warning setting tool (CRPT) in 3.1.1 Verification Mode in Consolidated Edition / Non-consolidated Edition (Import of Initial Quantity: No)."
Orange	Emergency Warning	Parts corresponding to the "Remaining count urgent warning" of LWS screen are eligible. For the setting method, refer to the instruction manual of the LNB. (In CM/DT line, the urgent warning message is not displayed.)
Red	Part Exhaust	

4.2 Work Progress Monitor Screen

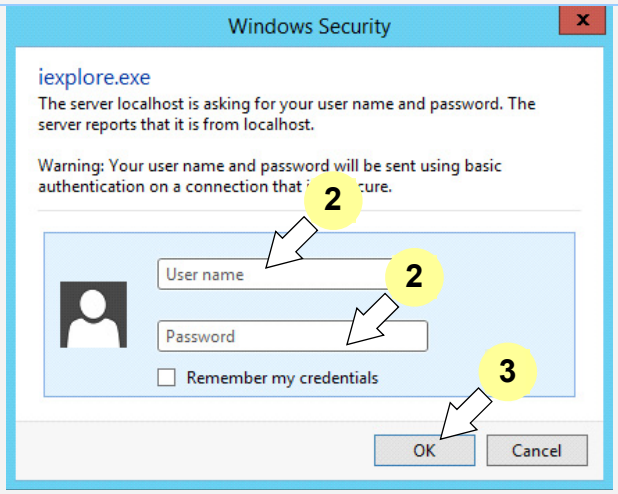
1
Log in to the following URL of PanaCIM Application server.

http://(PanaCIMApplicationServerIP address)/eplantview



2
When the login screen appears, enter your user name and password.

3
Click the [OK].



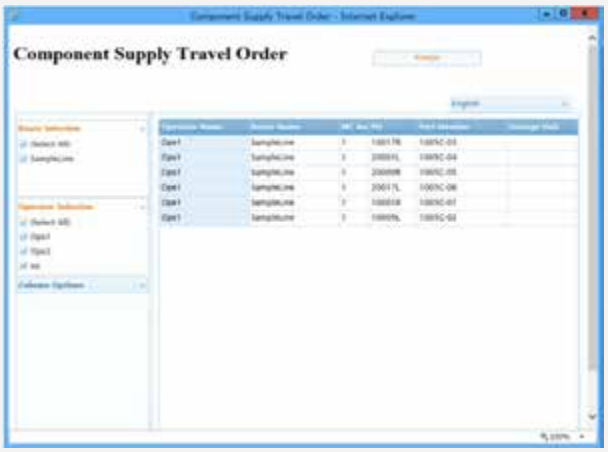
4
When the menu screen appears, Select [Production Monitoring&Dispatch]-[Travel Order].



5
Component Supply Travel Order screen appears.

* When the size of the display window is small, you may horizontal scroll bar is not displayed.
To display the horizontal scroll bar, you have the following methods.
- The increasing size of the display window.
- Use the vertical scroll bar send the screen to the bottom.

* Update cycle of the screen is 60 seconds.
However, when there is no change in the travel order information, it does not change the appearance on the screen.



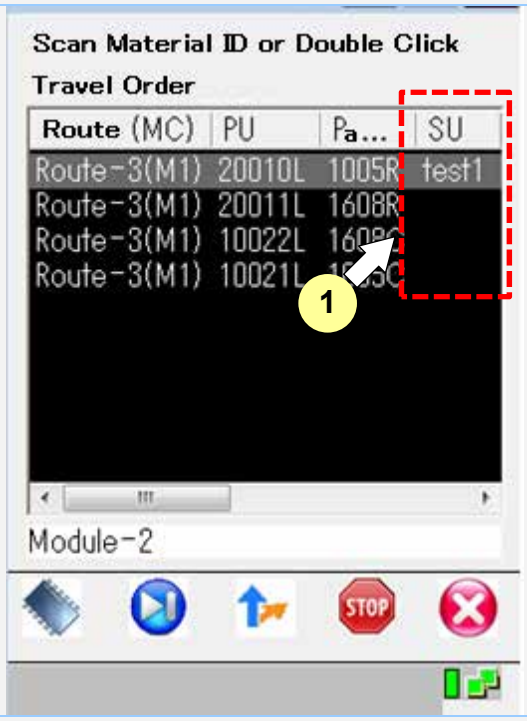
5. Manage of In-line Shelf

This feature can be activated only with Material Verification (Basic) and Material Management (Reservation) license.

5.1 Show Part Location

Parts are monitored individually using the material management module. When storage unit is linked with machines as explained in 2.4, the in-line parts shelf information appears on Travel Order screen.

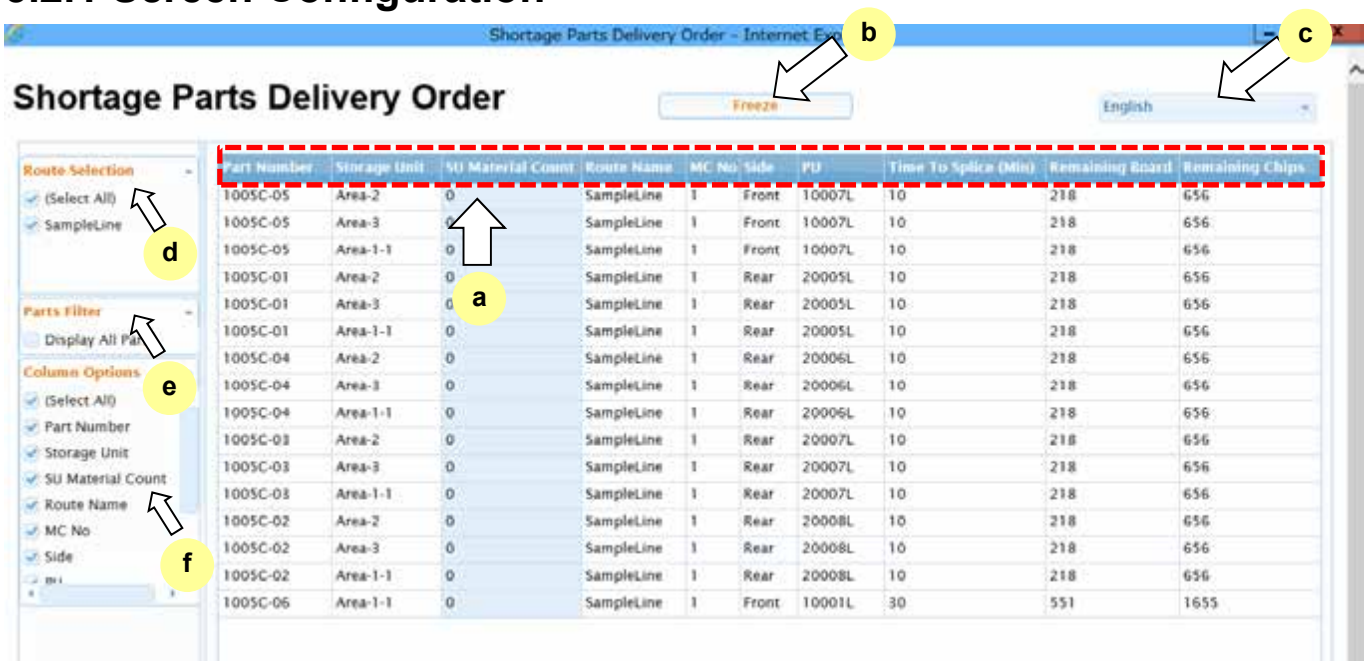
1
Line side shelf (storage unit) information appears in the column SU on Travel Order screen.
When a part is in a storage unit, the unit name appears. (If a part is stored in multiple units, the name of the nearest one appears.)



5.2 Delivery Order to In-line Part Shelf

Parts are monitored individually using the material management module. When storage unit is linked with machines as explained in 2.4, the in-line parts shelf information appears on Travel Order screen.

5.2.1 Screen Configuration



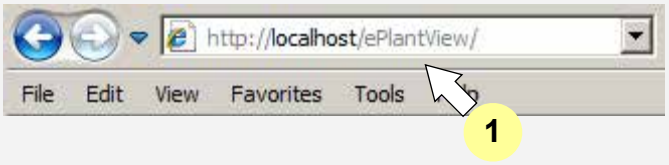
No.	Item	Details
	Part Number	The number of supplying part
	Storage Unit	The name of storage unit where supplying parts are
	SU Material Count	The number of reels/trays in the storage unit
	Route Name	The name of route (line) where parts are supplied
	Machine Number	Supply target machine number
	Side	Show the supply position Front or Rear
a	PU	Feeder address for each digit, it becomes the following content. X***** : Table number *YYYY* : Feeder address *****Z : Feeder sub address Feeder address will be the following. In the case of tape feeder : 0001-0030 In the case of single tray or twin tray on the left side : 1001-1020 In the case of twin tray : 2001-2020 Feeder sub address is the following. Feeder left (in the case of a double feeder / thin type tape feeder), or in the case of a single feeder: “L” Feeder right (in the case of a double feeder / thin type tape feeder), or in the case of a single feeder: “R” In the case of tray: empty

	Time to Splice (Min)	Remaining time until splicing limit length *The default splicing length limit value is 1,200mm, and for LNB, it is alterable with LWS (in CM/DT line, it is not alterable). * A formula: Chip number of remaining until the splicing limit length * cycle time / pickup count of times per sheet ". "Cycle time" and " pickup count" is calculated from the data in the actual production.
	Remaining Board	The number of boards that can be produced before part exhaust
	Remaining Parts	Remaining number of parts being used in the on-going production
b	Freeze/Update	[Freeze] stops updating the information, and [Update] to start updating
c	Language	Select a language for the display from Japanese, English and Chinese.
d	Route Selection	Select a route to display
e	Parts Filter	Check [Display All Parts] to display all parts. Uncheck it to show only the parts that are not in the storage unit.
f	Column Option	Select column(s) to display

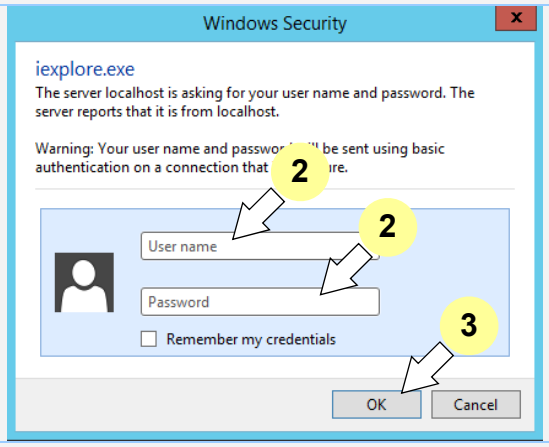
5.2.2 Shortage Parts Delivery Order Screen Display

1
Log in to the following URL of PanaCIM Application server.

http:///(PanaCIMApplicationServerIP address)/eplantview



2
Enter login user name and password on the login screen.

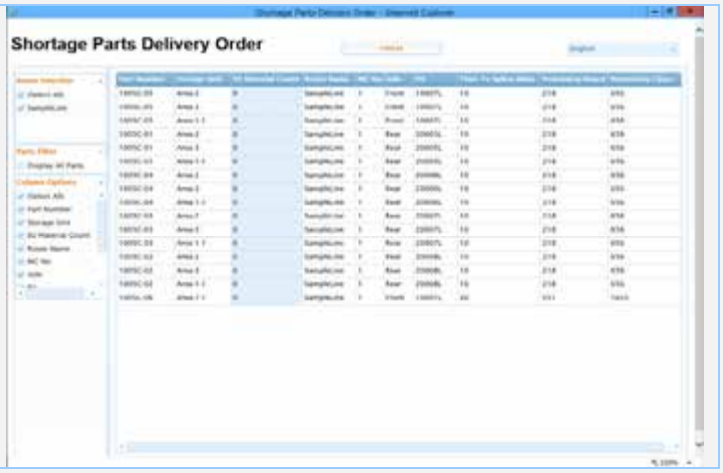


3
Click [OK].

4
Select [Production Monitoring&Dispatch]-[Shortage Parts Delivery Order].



5
Shortage Parts Delivery Order screen appears.



* When the size of the display window is small, you may horizontal scroll bar is not displayed.
To display the horizontal scroll bar below.
・To increase the size of the display window.
・Use the vertical scroll bar send the screen to the bottom.

* Update cycle of the screen is 60 seconds.
However, when there is no change in the instruction information, it does not change the appearance on the screen.

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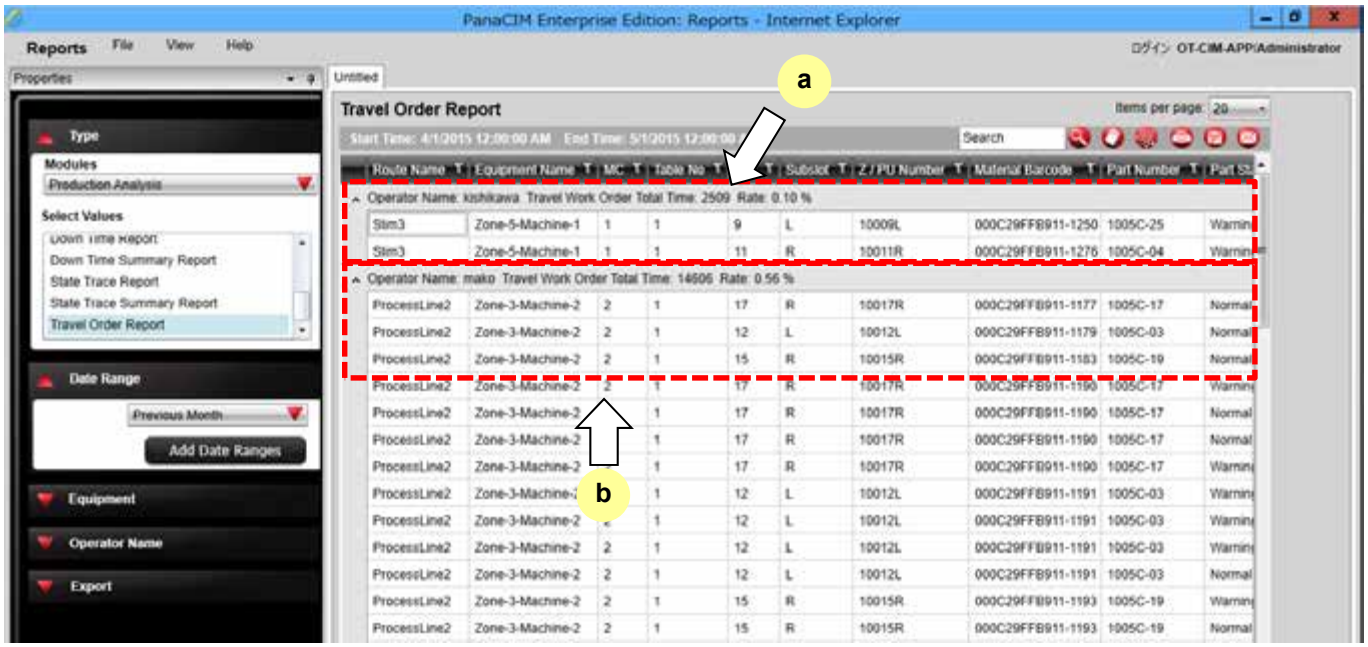
6. Manage of Travel Order report

This feature is available only with Production Analysis License.

6.1 Manage of Travel Order report

Time and other details of parts supply work are displayed for each operator.

6.1.1 Screen Configuration

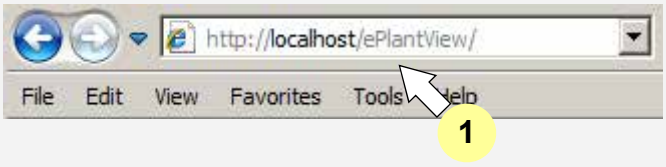


No.	Item	Details
a	Summary	Time summary per operator
	Operator Name	The target operator name
	Travel Work Order Total Time (Sec)	Total work time within the specified data
	Rate	Work time rate within the selected period
b	Detail	Detailed information per parts supply work
	Line Name	The name of the line where parts were supplied
	Equipment Name	The name of the equipment where parts were supplied
	MC	The number of the machine where parts were supplied
	Table Number	The number of the table where parts were supplied
	Slot Number	The number of the slot where parts were supplied
	Sub-slot Number	The number of the sub-slot where parts were supplied
	Z/PU Number	The feeder address where parts were supplied
	Material Barcode	The part barcode of the supplied parts.
	Part Number	The part number of the supplied parts.
	Part Status	Parts status, emergency warning/warning/part exhaust at the time of supply
	Work Time (sec)	Time required for the supply
	Time Stamp	Time when parts were supplied

6.1.2 Open Travel Order Report

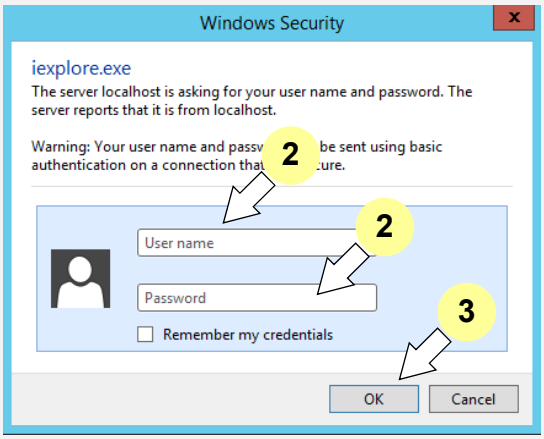
1
Log in to the following URL of PanaCIM Application server.

http://(PanaCIMApplicationServerIP address)/eplantview



2
Enter login user name and password on the login screen.,

3
Click [OK].



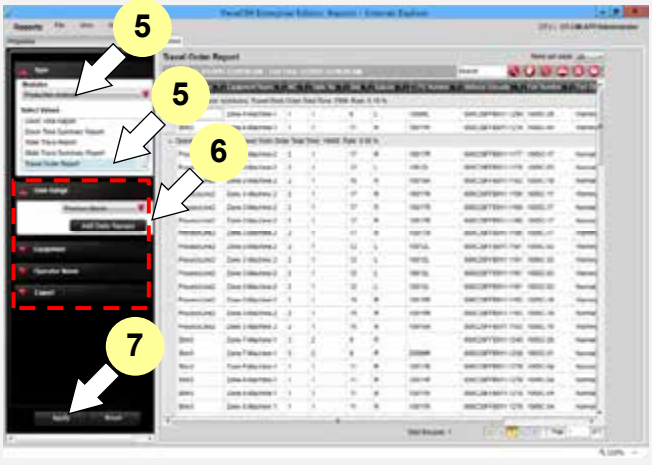
4
Select [Chart and Reports]–[Reports].



5
Select below from the menu.
Module: Production Analysis
Select Value: Travel Order Report

6
Select [Date], [Machine] and [Operator Name] for data display.

7
Click [Apply] and Travel Order Report appears.



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Panasonic Smart Factory Solutions Co., Ltd.

4-1-62 Minoshima, Hakata-ku, Fukuoka City, Fukuoka, Japan
Phone number: +81-92-477-1205
Fax number: +81-92-477-1664

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